System Relaxation for Resilient Network Governance: A Model-Driven Control Mechanism Against Structural Stress Concentration

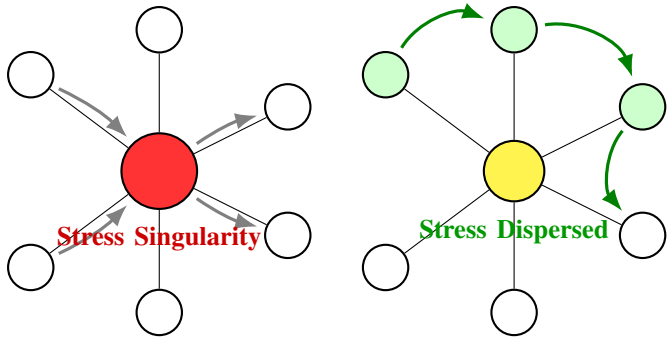
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***Abstract*—Prevailing network control strategies, rooted in static shortest-path logic, inherently suffer from "stress concentration," where traffic excessively gravitates toward critical nodes, causing structural vulnerabilities and premature capacity saturation. While existing load-aware or AI-driven methods attempt to mitigate congestion through reactive pathfinding, they fail to address the underlying structural cause by neglecting to proactively control the network's cost landscape. To address this, we introduce the System Relaxation Algorithm (SRA), a model-driven supervisory governance layer inspired by physical relaxation that guides the network toward a "relaxed" emergent equilibrium of load balance. SRA functions as an interpretable "white-box" dynamical system where node costs evolve based on historically accumulated structural stress (both topological betweenness centrality and generalized demand- and capacity-weighted stress). By modeling discrete path switching as a non-smooth dynamical system, we provide rigorous proofs of global convergence to a compact invariant attracting set and practical stability using piecewise Lyapunov functions and minimum dwell-time regularization, revealing that traditional reactive routing is a degenerate, memoryless case of this broader framework where system damping vanishes. Extensive simulations across synthetic (BA, WS, ER up to $N = 500$) and real-world backbone topologies (NSFNET, GEANT) demonstrate that SRA's behavior is profoundly topology-dependent: in heterogeneous networks, it acts as a proactive performance optimizer, reducing peak centrality by over 80% and increasing throughput by more than 45% (up to 107.4% at $N = 500$ while suppressing route-flapping frequency by 79.7%); in homogeneous topologies, it intelligently shifts its objective to resilience enhancement. SRA thus establishes a predictable, mathematically tractable paradigm for network governance that achieves an intelligent, topology-aware trade-off between performance and resilience.**

***Index Terms*—Network control, stress-aware governance, network resilience, self-organizing systems, load balancing, non-smooth dynamical systems, stability analysis.**

I. Introduction

MODERN communication networks, from data center fabrics to large-scale ISP backbones, operate under increasingly volatile traffic conditions where stability is paramount. In such high-stakes environments, prevailing network control paradigms rooted in static shortest-path logic face a fundamental crisis. These traditional protocols inherently lack a mechanism for managing structural stress, leading to a phenomenon we term "stress concentration"—a severe imbalance where load disproportionately gravitates toward a few critical topological bottlenecks. This is not merely a transient congestion issue but a structural vulnerability: it creates "stress singularities" that render the network fragile to cascading failures and causes premature capacity saturation, where bottlenecks induce massive packet loss and soaring latency long before the network's physical capacity is reached.



(a) Conventional Paradigm: Stress Concentration

(b) SRA Control Paradigm: System Relaxation

Fig. 1: Conceptual illustration of the core problem and the proposed SRA paradigm. SRA mitigates the "stress concentration" inherent in conventional network control by guiding the system to a "relaxed" state of balanced load.

The root of this fragility lies in a flawed focus on reactive symptoms rather than proactive structural governance. Prior work has largely attempted to patch this issue through three dominant approaches, yet all fall short of addressing the underlying cause:

- **Reactive Load-Aware Routing (e.g., [1]):** These methods adjust paths based on transient metrics like queue length. They are essentially *symptomatic treatments* that oscillate under high load due to delayed feedback loops.
- **Static Centrality & IGP Weight Optimization (e.g., [2], [3]):** Offline link-weight tuning and static centrality-aware routing optimize weights for an average snapshot in an open-loop manner or via local search without damped state memory, causing route flapping when applied iteratively under discrete path switching.
- **Black-box AI Control (e.g., [4]):** While Deep Reinforcement Learning (DRL) agents can implicitly learn

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topology features, they function as uninterpretable "black boxes," lacking the safety guarantees required for mission-critical infrastructure.

Explicit Research Gap: As summarized in Table I, no existing framework simultaneously provides (i) *closed-loop structural stress governance* that reshapes the routing cost landscape before queue overflows occur, (ii) *non-smooth Lyapunov stability and anti-chattering guarantees* under discontinuous shortest-path switching, and (iii) *white-box analytical tractability* linking topological peak shaving to network capacity expansion.

To bridge this gap, we propose a paradigm shift from reactive pathfinding to proactive *stress governance*, inspired by the concept of "relaxation" in physical systems. We introduce the System Relaxation Algorithm (SRA), a model-driven control system designed to disperse structural stress by dynamically reshaping the network's cost landscape. Unlike black-box approaches, SRA operates as an interpretable, white-box dynamical system: it employs a "leaky integrator" mechanism coupled with minimum dwell-time temporal regularization to accumulate historical stress into a controllable state variable, thereby guiding the network toward an emergent equilibrium of load balance (as conceptually illustrated in Fig. 1). Crucially, this control paradigm aligns perfectly with Software-Defined Networking (SDN) architectures, leveraging the controller's global view to calculate stress and its centralized authority to enforce the resulting cost policies.

Our main contributions are summarized as follows:

1) **A Unified Control-Theoretic Framework:** We formalize structural stress—both unweighted topological betweenness centrality and generalized demand- and capacity-weighted stress $\widetilde{\kappa}_i(S; W, \mu)$—as a controllable state variable within a closed-loop dynamical system. **This framework explicitly decouples structural stress governance from low-level pathfinding**, revealing that conventional reactive routing corresponds to the degenerate memoryless limit ($\alpha = 1$) where system damping vanishes.
2) **Rigorous Analysis of Discontinuous Dynamics and Capacity Expansion:** We model discrete path switching in centrality-based routing as a *non-smooth dynamical system* via Krasovskii differential inclusions [6]. Using piecewise Lyapunov functions and minimum dwell-time regularization (T_d), we establish (i) global convergence to a compact invariant attracting set Ω, (ii) the *Dissipation-Injection Law* bounding steady-state oscillations, and (iii) formal peak-shaving and capacity-expansion conditions (Lemma 2 and Proposition 3) linking structural stress dispersion to network throughput gains.
3) **Comprehensive Empirical Validation Across Synthetic and Real-World Topologies:** Through extensive multi-seed simulations across BA, WS, and ER networks ($N = 50$ to 500) as well as the real-world NSFNET and GEANT backbones, supplemented by component-wise ablation studies, Fortz–Thorup IGP weight-tuning comparisons, skewed Pareto gravity traffic, time-varying load surges, and sensing noise injection, we demonstrate SRA's emergent topology-aware duality and scalability.

This work focuses on the algorithmic foundations and stability analysis of stress-aware network governance. While we discuss the alignment with SDN architectures, specific hardware implementations and south-bound protocol extensions are beyond the scope of this paper. The remainder of this paper is organized as follows: Section II categorizes the existing literature and positions SRA within the routing landscape. Section III establishes the unified theoretical framework, detailing the control-theoretic model, the rigorous stability analysis via non-smooth dynamics, and the formal bridge to capacity expansion. Section IV presents the empirical validation and analysis across the topological spectrum, and Section V concludes the paper.

II. Related Work

The mitigation of structural stress remains a fundamental challenge at the intersection of network science and traffic engineering. While existing methodologies strive to balance traffic distribution, they vary significantly in their perception of network states and their subsequent control mechanisms. As synthesized in Table I, there exists a discernible gap between traditional path-finding strategies and the active reshaping of a network's cost landscape. To situate the proposed System Relaxation Algorithm (SRA) within the broader research context, we categorize the existing literature into three distinct paradigms: reactive symptomatic management, static structural optimization, and black-box learning control.

A. Reactive Control: Symptomatic Relief and Its Limitations

The most pervasive paradigm in network control relies on *reactive* feedback loops that adjust paths based on real-time performance metrics. These protocols typically monitor local queue lengths [1], energy-aware load indicators [7], or composite link quality in highly dynamic satellite and vehicular environments [8], [9]. For instance, the CRADLE protocol [1] leverages cross-layer cues to deflect traffic, while multi-criteria decision models [10] and hierarchical cluster-based scaling [11] attempt to optimize routing decisions under fluctuating demands. Recent explorations in IoT and smart cities further extend this logic to position-based and occlusion-driven routing [12], [13], addressing delays that often cascade across high-speed transport nodes [14].

While effective at mitigating immediate congestion, these methods essentially function as "symptom relievers" that address buffer overflows or link saturation rather than the underlying *structural root causes*. By treating the network's cost landscape as an immutable constraint, reactive paradigms remain passive followers of instantaneous traffic noise. Consequently, they frequently suffer from the "ping-pong" effect—an instability where paths oscillate between overloaded and underloaded states due to delayed feedback. Lacking a long-term integration mechanism like SRA's leaky integrator, these approaches fail to distinguish between transient traffic spikes and endemic structural stress, and they lack the rigorous mathematical framework necessary to guarantee global stability under non-smooth traffic dynamics.

TABLE I: Taxonomy and Comparative Analysis of Network Control Paradigms

Feature	**Reactive Queue-Aware [1]**	**IGP Weight Tuning [2]**	**Static Centrality [3]**	**AI-Driven / DRL [4], [5]**	**SRA (This Work)**
Sensing Object	Instantaneous queue length / delay	Link utilization ratio (l_e/c_e)	Static topological BC ($\kappa_i^{(0)}$)	High-dimensional state embeddings	**Accumulated structural stress** ($S_i^{(k)}$)
Control Law	Memoryless reactive deflection ($\alpha = 1$)	Local search / convex utilization penalty	Open-loop one-shot static cost penalty	Black-box neural policy (DRL/GNN)	**Leaky integrator** ($\alpha \in (0,1)$) **+ Dwell time** T_d
Mathematical Basis	Heuristic cost functions	Mixed-integer / local search heuristic	Static graph theory	Empirical reward maximization	**Non-smooth differential inclusions [6]**
Stability Guarantee	None (prone to route flapping)	None under online closed-loop switching	N/A (Open-loop static)	None (no formal Lyapunov guarantee)	**Global attracting set** Ω **& Drop-Jump Law**
Anti-Chattering	None ($\sim 98\%$ switch rate)	None (continuous weight updates)	Static (cannot adapt)	Heuristic reward shaping	**Provable bound** $\leq 1/T_d$ **(Theorem 4)**
Per-Step Complexity	$O(E + N \log N)$	$O(I_{\text{LS}} \cdot N(E + N \log N))$	$O(N(E + N \log N))$ (once)	$O(L_{\text{GNN}} E d^2)$ + offline training	$O(N(E + N \log N))$ **centralized /** $O(1)$ **dist.**
Interpretability	High	Medium	High	Low (Black-box)	**High (White-box Control Law)**

B. Static Structural and IGP Weight Optimization: The Open-Loop Governance Gap

Recognizing the decisive role of network topology, a significant body of research leverages link-weight tuning [2] and centrality metrics to guide routing and resource allocation. Classical IGP weight optimization [2] adjusts OSPF link weights using piecewise-linear convex congestion penalties via local search. Betweenness Centrality (BC) [15] and its sophisticated variants, such as link-variance optimization for optical networks [3] and weighted gravity models for multiplex systems [16], represent the established methodologies for identifying topological bottlenecks and influential nodes [17], [18]. Theoretical advancements have further underscored that routing via quasi-shortest paths (ρ-geodesics) [19] or optimizing connectivity under probabilistic failures [20] can significantly enhance network resilience. To circumvent the $O(N^3)$ computational complexity inherent in global centrality analysis, recent literature has proposed efficient approximation techniques, including hyperbolic space embeddings [21], clustered parallelization [22], and distributed estimation frameworks [23], [24]. These structural insights have been applied across diverse domains, from optimizing SDN flow table storage [25] to enabling greedy forwarding in sparse networks [26] and energy-efficient path-finding in sensor networks [27].

Despite their analytical depth, these structural optimization methods are fundamentally *open-loop* and descriptive in nature. Even when accounting for temporal variations in centrality [28], the metrics are treated as passive observations rather than controllable state variables. When static centrality penalties or unregularized weight updates are applied repeatedly in closed loop, discrete shortest-path switching induces severe route oscillations. Unlike SRA, these approaches lack a rigorous control-theoretic feedback law (combining leaky integration with dwell-time regularization) to map structural observations into a provably stable cost landscape.

C. AI-Driven and Self-Organizing Control: The Black-Box Dilemma

The recent surge in machine learning has catalyzed the deployment of Deep Reinforcement Learning (DRL) and Graph Neural Networks (GNNs) for complex network management. State-of-the-art agents, such as GRL-PS [4] and multi-agent GNN frameworks [5], attempt to implicitly capture topological features to navigate intricate congestion scenarios. To mitigate the computational overhead of global metrics, some approaches utilize node embeddings to approximate traditional centrality measures [29]. Parallel to these learning-based methods, bio-inspired self-organizing models [30]–[32] and heuristic search strategies—including discrete particle swarm optimization [33] and genetic algorithms [34]—have been applied to dynamic cluster-based routing. These autonomous path-finding capabilities are particularly explored in mission-critical environments such as missile swarm networks [35], where nodes must adapt to rapid topology changes.

Despite their empirical performance, these approaches frequently operate as *black boxes* with opaque decision logic. They generally lack formal safety verification or rigorous Lyapunov stability guarantees, posing systemic risks to critical infrastructure where the unpredictable convergence of DRL agents remains a significant concern. Furthermore, heuristic and self-organizing models often overlook the non-smooth discontinuities inherent in discrete path-switching actions. This theoretical void is bridged by SRA, which provides a "white-box" alternative through the rigorous application of non-smooth dynamical systems theory [6], ensuring both mathematical explainability and guaranteed global convergence.

In summary, while the aforementioned paradigms have significantly advanced network engineering, a fundamental gap persists between topological awareness and active governance. Reactive methods prioritize speed over stability, static structural optimizations offer diagnostic depth but lack execution mechanisms, and learning-based controls trade mathematical explainability for empirical performance. Specifically, the networking community still lacks a unified framework that can

bridge the "governance gap" between high-level structural observations (centrality) and low-level traffic control laws with guaranteed global stability. This paper addresses these limitations by introducing the System Relaxation Algorithm (SRA). By grounding the control logic in non-smooth dynamical systems theory and the "Cost Law" mechanism, SRA transitions the network management philosophy from passive path-finding to the active, stable, and explainable reshaping of the network's cost landscape.

III. Theoretical Model and Methods

We formulate the problem of stress-aware network governance not as a sequence of heuristic routing updates, but as a **discrete-time, non-smooth dynamical system**. This control-theoretic perspective is decisive: it allows us to transcend the stability pitfalls of naive reactive protocols, which often collapse into destructive oscillations due to their lack of system memory. In our framework, the network's cost landscape is no longer a static input, but a dynamic state variable governed by a physics-inspired "leaky integrator." This section provides a systematic construction of this system. We first define the network model and formalize structural stress as a controllable state. We then detail the core state evolution law that couples continuous stress accumulation with discrete path switching. The culmination is a rigorous analysis using piecewise Lyapunov functions and discrete-time inclusion theory, proving that SRA guarantees global convergence to a robust equilibrium—a theoretical guarantee that heuristic methods fundamentally lack.

A. Network Model, Generalized Stress Metric, and Problem Formulation

We model the underlying network topology as a connected undirected graph $G = (V, E)$ with $N = |V| \geq 3$ nodes. From a control perspective, the network's "plant" is governed by a dynamically evolving node cost vector $C \in \mathbb{R}_+^N$. This vector defines the instantaneous *cost landscape*, where the cost of any path P is the sum of the transit costs of its constituent nodes, i.e., $d(P) = \sum_{v \in P} C_v$. Traffic flows are assumed to follow lowest-cost paths within this landscape.

To ensure mathematical transparency, our control model operates under four explicit structural and operational assumptions:

- **Assumption 1 (2-Vertex-Connected Core / Alternative Paths):** For bottleneck nodes of interest, the graph G contains at least two vertex-disjoint simple paths between major source-destination pairs (for true cut-vertices or bridges, structural stress is physically irreducible; see Section V-B).
- **Assumption 2 (Time-Scale Separation):** Control-plane cost updates occur at discrete epochs $k \in \mathbb{N}$ on a slower time scale than data-plane packet forwarding, allowing path-level flow rates to reach quasi-stationary equilibrium within each control epoch.
- **Assumption 3 (Deterministic Tie-Breaking):** Equal-cost shortest paths are either split equally (ECMP) or resolved via a fixed lexicographical tie-breaking rule, guaranteeing that the set of realizable shortest-path configurations $\mathcal{P}$ on any finite graph G is finite ($|\mathcal{P}| = M < \infty$).
- **Assumption 4 (Quasi-Stationary Demand and Service Capacity):** Over a relaxation horizon, each origin-destination (OD) pair (s, t) has a non-negative relative traffic weight $W_{st} \geq 0$, and each node $i \in V$ has an effective service capacity $\mu_i > 0$.

To quantify the structural stress induced by a specific cost distribution, we employ **Betweenness Centrality (BC)** [15] as the baseline topological feedback signal. The unnormalized topological stress on node i, denoted $\kappa_i(S)$, represents its absolute path load under uniform unit demand ($W_{st} \equiv 1$):

$$\kappa_i(S) = \sum_{s \neq i \neq t \in V} \frac{\sigma_{st}^{C(S)}(i)}{\sigma_{st}^{C(S)}} \tag{1}$$

where $\sigma_{st}^{C(S)}$ is the total number of shortest paths between s and t under the cost configuration $C(S)$, and $\sigma_{st}^{C(S)}(i)$ is the subset passing through i.

Furthermore, to address non-uniform traffic matrices $W = [W_{st}]$ and heterogeneous node service capacities $\mu = (\mu_i)_{i \in V}$, we generalize Eq. (1) to the **Demand- and Capacity-Weighted Structural Stress**:

$$\widetilde{\kappa}_i(S; W, \mu) := \frac{1}{\mu_i} \sum_{s \neq i \neq t \in V} W_{st} \frac{\sigma_{st}^{C(S)}(i)}{\sigma_{st}^{C(S)}}. \tag{2}$$

When demands are uniform ($W_{st} \equiv 1$) and node capacities are homogeneous ($\mu_i \equiv \mu$), $\widetilde{\kappa}_i(S; W, \mu)$ reduces to $\kappa_i(S)/\mu$, whose max-normalization is identical to that of topological BC in Eq. (1). Crucially, because W_{st} and μ_i are constants within a control horizon, $\widetilde{\kappa}(S; W, \mu)$ remains piecewise constant over the same finite partition of the state space $X = [0, 1]^N$, so all subsequent stability and capacity proofs hold identically for both $\kappa(S)$ and $\widetilde{\kappa}(S; W, \mu)$.

Since $\kappa_i(S)$ is unbounded with network size, we project this feedback signal into a compact state space to ensure system stability. We define the max-normalized stress vector as:

$$\mathrm{norm}(\kappa(S)) := \begin{cases} \dfrac{\kappa(S)}{\max_j \kappa_j(S)}, & \text{if } \max_j \kappa_j(S) > 0, \\ \mathbf{0}, & \text{otherwise.} \end{cases} \tag{3}$$

This normalization confines the feedback signal strictly within the unit hypercube $[0, 1]^N$. **Key Geometric Property:** Crucially, for any finite graph with a deterministic tie-breaking rule, the set of all possible shortest-path configurations is finite. This implies that $\mathrm{norm}(\kappa(S))$ does not vary continuously but jumps between a **finite set of constant vectors** $\{v_1, \ldots, v_M\} \subset [0, 1]^N$. This quantization is non-trivial; it renders the system dynamics *piecewise constant*, forming the topological foundation for our non-smooth convergence analysis in Section III-C.

Formal Problem Formulation: Formally, given (G, W, μ), we seek a closed-loop state feedback policy mapping historical stress observations $\{\kappa^{(0)}, \ldots, \kappa^{(k)}\}$ to node transit costs

$C^{(k+1)} \in [1, 1+\beta_I]^N$ so as to solve the bi-criteria resilient governance problem:

$$\begin{aligned} \min_{\{S^{(k)}, C^{(k)}\}} \quad & \limsup_{k\to\infty} \max_{i\in V} \widetilde{\kappa}_i(S^{(k)}; W, \mu) + \eta_{\text{sw}}\, \bar{f}_{\text{sw}} \\ \text{s.t.} \quad & S^{(k)} \in [0,1]^N, \quad C^{(k)} = \mathbf{1} + \beta_I S^{(k)}, \\ & \bar{f}_{\text{sw}} := \limsup_{K\to\infty} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{I}\big(\mathcal{P}(C^{(k+1)}) \neq \mathcal{P}(C^{(k)})\big) \leq \frac{1}{T_d}, \end{aligned} \tag{4}$$

where $\mathcal{P}(C^{(k)})$ denotes the active shortest-path configuration at step k, $\bar{f}_{\text{sw}}$ is the asymptotic route-switching frequency, and $T_d \geq 1$ is the minimum dwell-time constraint.

B. SRA: Core Mechanism and Algorithm Implementation

Design Philosophy: SRA as a Supervisory Layer. Crucially, SRA is not a routing protocol designed for packet forwarding; rather, it functions as a supervisory governance layer that optimizes the network's "cost landscape." It provides an adaptive metric space upon which standard routing protocols (e.g., OSPF, IS-IS, or Dijkstra) can then execute without triggering structural bottlenecks.

Having established the state space, we now derive the *laws of motion* for the SRA controller. We formally architect SRA not merely as a routing algorithm, but as a **closed-loop feedback control system**, as visualized in Fig. 2. The central design philosophy is to decouple the instantaneous observation of stress from the immediate actuation of costs. By inserting a temporal integration stage—specifically, a **leaky integrator**—between sensing and actuation, the SRA controller governs the underlying network plant with a critical damping effect. This design fundamentally distinguishes SRA from "amnesic" reactive protocols, allowing it to filter out high-frequency transient noise and respond only to persistent structural trends.

A naive feedback mechanism that sets node costs directly proportional to instantaneous centrality is inherently unstable, creating a "bang-bang" control regime that leads to persistent, destructive oscillations. To ensure rigorous stability, SRA's design must incorporate system memory to dampen these high-frequency fluctuations. We achieve this using an Integral (I) control structure, implemented as a **leaky integrator**. We define a core state vector, the "accumulated pressure" $S \in \mathbb{R}_+^N$, which represents the historically integrated stress on each node. This state evolves according to the discrete-time transition equation:

$$S_i^{(k+1)} = (1-\alpha) S_i^{(k)} + \alpha \cdot \text{norm}(\kappa_i^{(k)}) \tag{5}$$

where k is the iteration step and the learning rate $\alpha \in (0,1)$ governs the system's memory persistence. Physically, this equation functions as a **digital low-pass filter**: the $(1-\alpha)$ term provides a critical **damping** effect, allowing the system to absorb transient traffic shocks while tracking persistent structural trends.

Remark 1 (Relationship to Reactive Routing)**.** It is critical to observe that standard reactive routing protocols correspond to a *degenerate case* of Eq. (5) where $\alpha = 1$.

In this memoryless limit, the damping term vanishes ($1-\alpha = 0$), explaining why such protocols inherently lack stability under high load.

The node transit cost C, the system's actuation signal, is then defined as an affine control law based on this stable historical state:

$$C_i^{(k+1)} = 1 + \beta_I \cdot S_i^{(k+1)} \tag{6}$$

where 1 is the base physical cost and $\beta_I > 0$ is the "pressure-to-cost" gain factor. By deriving costs from the system's history (S) rather than instantaneous noise, SRA makes stable, proactive decisions.

The complete iterative process is formalized in Algorithm 1, and its cyclical workflow is visualized in Fig. 3. To address the non-smooth nature of discrete path switching, the algorithm introduces a *stabilization constraint*: the minimum dwell time T_d. This mechanism prevents the system from reacting to the immediate perturbations caused by its own control actions, allowing the network state to settle before the next update. The core principle of SRA is to fundamentally reshape the network's "cost landscape" based on accumulated structural stress. As shown conceptually in Fig. 4, this forces the routing protocol to intelligently trade path efficiency for bottleneck avoidance, rerouting traffic from a physically short but high-stress path to a longer but more stable peripheral route.

Complexity and Scalability Analysis

The computational complexity of the baseline SRA iteration is governed by the exact centrality calculation, scaling as $O(N(E + N\log N))$ time and $O(N+E)$ memory using Brandes' algorithm [15]. Moreover, when the dwell-time constraint $T_d > 1$ is active, shortest-path recomputation is skipped during the $T_d - 1$ intra-block cooling steps, reducing the amortized per-epoch time complexity by a factor of $1/T_d$ during switching transients. While this overhead is negligible in centralized SDN deployments due to the separation of control-plane computation and data-plane forwarding (e.g., ~ 31 ms per iteration for $N = 100$ and ~ 851 ms for $N = 500$; see Section IV-H), SRA also supports a **distributed sensing mode** for massive-scale networks. In this mode, we relax the requirement for exact global calculation. Instead, individual nodes act as distributed estimators: they approximate their own local centrality by tracking the cardinality of unique source-destination pairs (or flow bytes weighted by $1/\mu_i$) in transiting flows [23]. These local stress indicators are then efficiently aggregated via standard management protocols (e.g., OSPF-TE extensions or SDN In-band Network Telemetry), effectively reducing the per-node computational complexity to $O(1)$. This hybrid architecture ensures that SRA remains scalable and deployable, transitioning seamlessly from exact structural analysis in small/medium networks to statistical approximation in hyper-scale environments.

C. Theoretical Analysis of Model Convergence

We now establish the rigorous mathematical foundation for the SRA control loop. The central analytical challenge lies in the system's hybrid nature: the continuous evolution of

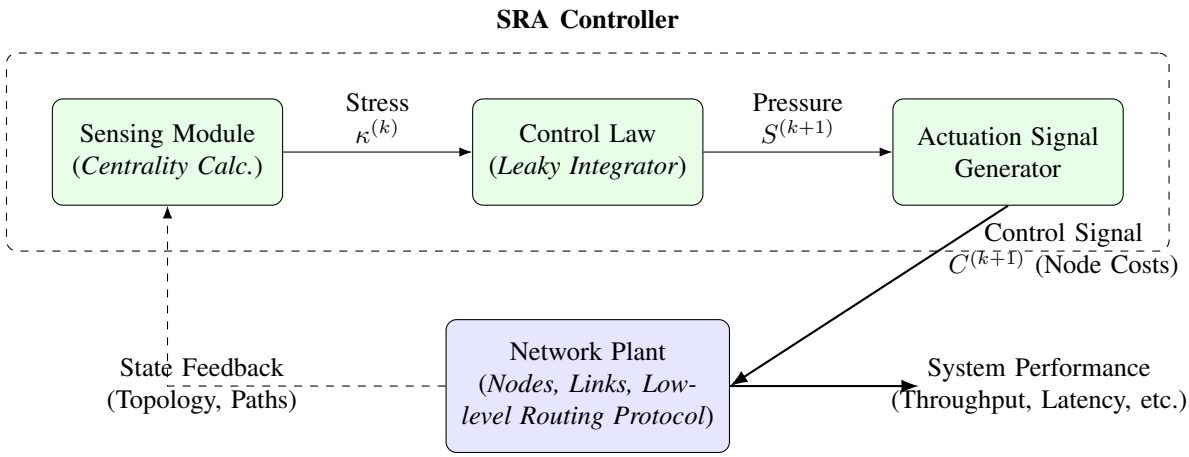


Fig. 2: The architecture of the SRA as a closed-loop control system. The SRA controller acts as a supervisory layer that governs the underlying network plant by dynamically adjusting the cost landscape.

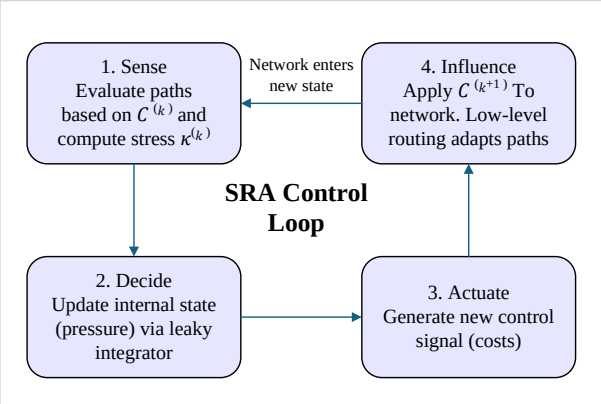


Fig. 3: The iterative workflow of the SRA control loop, illustrating the four key phases of sensing, decision-making, actuation, and system influence that drive the network toward a state of dispersed structural stress.

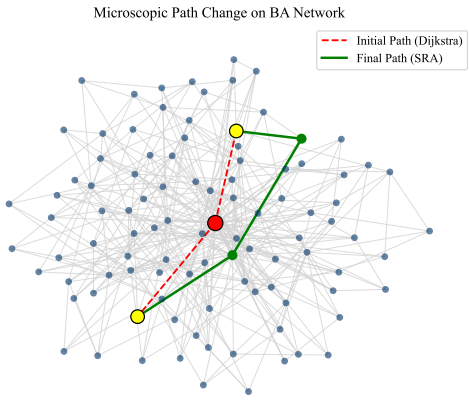


Fig. 4: A conceptual visualization of SRA's path reshaping mechanism. The algorithm diverts traffic from the efficient but congested 2-hop path (red dashed) through the central hub to a longer but more stable 4-hop peripheral path (green solid).

pressure states (S) is coupled with the discrete, discontinuous switching of routing topologies. This non-smoothness renders standard gradient-based control theory inapplicable and represents the primary theoretical barrier where heuristic protocols fail to guarantee stability.

Algorithm 1: System Relaxation Algorithm (SRA)

Input: Graph $G = (V, E)$; optional demand weights $W = [W_{st}]$ and node service rates $\mu = (\mu_i)$; learning rate $\alpha \in (0, 1)$; pressure-cost factor $\beta_I > 0$; stopping threshold $\varepsilon > 0$; max iterations $K_{\max}$; min dwell time $T_d \geq 1$.

Output: Final node cost vector $C^{(k+1)}$ and the corresponding routing policy $\mathcal{P}^{(k+1)}$.

Initialize: $S^{(0)} \leftarrow \mathbf{0}$; $C^{(0)} \leftarrow \mathbf{1}$; $\tau_{\text{dwell}} \leftarrow 0$

for $k = 0, 1, 2, \ldots, K_{\max} - 1$ **do**

`// Step 1: Path Evaluation with Dwell-Time Gating`

  if $\tau_{dwell} == 0$ **then**

    Compute all-pairs shortest paths $\mathcal{P}^{(k)}$ using node costs $C^{(k)}$

    if $k > 0$ ***and*** $\mathcal{P}^{(k)} \neq \mathcal{P}^{(k-1)}$ ***and*** $T_d > 1$ **then**

      $\tau_{\text{dwell}} \leftarrow T_d - 1$

  else

    Freeze path set $\mathcal{P}^{(k)} \leftarrow \mathcal{P}^{(k-1)}$ and $\tau_{\text{dwell}} \leftarrow \tau_{\text{dwell}} - 1$

`// Step 2: Structural Stress Calculation`

  Compute $\kappa^{(k)}$ (Eq. (1) or generalized $\widetilde{\kappa}^{(k)}$ in Eq. (2)) over $\mathcal{P}^{(k)}$, and set $q^{(k)} \leftarrow \text{norm}(\kappa^{(k)})$

`// Step 3: Leaky-Integrator Pressure Update`

  $S^{(k+1)} \leftarrow (1 - \alpha)S^{(k)} + \alpha\, q^{(k)}$

`// Step 4: Affine Cost Actuation`

  $C^{(k+1)} \leftarrow \mathbf{1} + \beta_I\, S^{(k+1)}$

`// Step 5: Convergence Check`

  if $\|S^{(k+1)} - S^{(k)}\|_\infty \leq \varepsilon$ ***and*** $\mathcal{P}^{(k)} == \mathcal{P}^{(k-1)}$ **then**

    break

To overcome this, we model the system dynamics using the formalism of discrete-time differential inclusions [6]. Our analysis proceeds in three logical steps to construct a complete stability proof:

1) **Global Attractivity (Safety):** We first prove the existence of a compact global attracting invariant set $\Omega \subset X$, guaranteeing that system trajectories are strictly confined within a bounded safety region and never diverge.
2) **Lyapunov Decay (Energy):** We construct a specialized *piecewise Lyapunov tracking energy* to quantify the system's "structural potential energy," proving that the feedback loop dissipates stress according to a strict dissipation-injection law.
3) **Practical Stability (Engineering):** Finally, we formalize the minimum dwell time T_d not as a heuristic parameter, but as a necessary condition to eliminate high-frequency chattering (Zeno-like switching), deriving an explicit convergence rate to a stable ε-neighborhood.

D. Global Attractiveness and Existence Guarantee for Convergence

The convergence analysis of centrality-based routing is notoriously difficult due to the piecewise constant, non-smooth nature of the map $\kappa(S)$. Heuristic approaches often ignore the discontinuities caused by path switching, leading to theoretical vacuums. To establish a rigorous guarantee, we do not merely approximate the system; we **regularize** the hybrid dynamics into a **discrete-time differential inclusion** [6]. The update rule is formalized as the set-valued map:

$$S^{(k+1)} \in F(S^{(k)}) := (1-\alpha)S^{(k)} + \alpha\,\mathcal{N}(S^{(k)}) \tag{7}$$

where $\mathcal{N}(S) := \bigcap_{\delta>0} \overline{\text{co}}\big(\text{norm}(\kappa(B_\delta(S)))\big)$ is the **Krasovskii regularization** of the discontinuous mapping $\text{norm}(\kappa(\cdot))$. Geometrically, this operation takes the closed convex hull of all limit points in a vanishing neighborhood of S, effectively "filling in the gaps" caused by discrete routing decisions (e.g., tie-breaking boundaries). This yields a mathematically well-behaved map, allowing us to derive the following foundational properties.

Lemma 1 (Global Confinement and Safety)**.** *Let the state space be the unit hypercube $X = [0,1]^N$. The discrete-time inclusion for SRA, given by $S^{(k+1)} \in F(S^{(k)})$, is strictly confined within X. Formally, X is a **compact, positively invariant set** ($F(X) \subset X$) and serves as an exponentially attracting compact absorbing set for any initial state $S^{(0)} \in \mathbb{R}^N$. Physical Interpretation: This lemma guarantees that the structural stress never diverges to infinity; the system possesses an intrinsic "safety boundary" dictated by the normalization mechanism.*

Proof Sketch: The proof demonstrates that if a state $S^{(k)}$ resides within the hypercube X, its successor $S^{(k+1)}$—being a convex combination of elements in X—must inherently remain within X. Moreover, for any out-of-bounds initialization $S^{(0)} \notin X$, the ℓ_∞ distance to X contracts geometrically by $(1-\alpha)$ at every step. *The full formal proof is provided in Appendix A.* ■

With safety guaranteed, we verify the topological regularity of the control map.

Proposition 1 (Topological Regularity)**.** *The set-valued map $F(\cdot)$ defined in (7) is **Upper Semi-Continuous (USC)** with **non-empty, compact, and convex** values.*

Proof Sketch: The proof exploits the piecewise constant structure of $\mathcal{N}(S)$, which stems from the finite nature of the network's path configurations (Assumption 3). This structure ensures that the graph of the map is closed, a sufficient condition for the USC property in compact spaces. *The full formal proof is provided in Appendix B.* ■

Combining the global confinement (Lemma 1) with topological regularity (Proposition 1), we establish the system's ultimate destiny.

Theorem 1 (Global Convergence to a Compact Invariant Attracting Set)**.** *For the discrete-time inclusion (7), the nested intersection*

$$\Omega := \bigcap_{n=0}^{\infty} F^n(X) \subset X := [0,1]^N \tag{8}$$

*is a **non-empty, compact, and positively invariant global attracting set** ($F(\Omega) = \Omega$), such that for any initial configuration $S^{(0)} \in X$:*

$$\text{dist}\big(S^{(k)}, \Omega\big) \;\to\; 0 \qquad (k \to \infty). \tag{9}$$

Proof Sketch: Because X is compact and positively invariant ($F(X) \subset X$) and F is USC with compact values (Proposition 1), the forward images $K_n := F^n(X)$ form a nested sequence of non-empty compact subsets $K_{n+1} \subset K_n$. By Cantor's intersection theorem, $\Omega = \bigcap_{n\geq 0} K_n$ is non-empty, compact, and invariant, and attracts all trajectories initialized in X. *The full formal proof is provided in Appendix C.* ■

Remark 2 (Attraction to a Limit Set vs. Unique Point Equilibrium)**.** We emphasize that Theorem 1 guarantees global convergence to a *compact invariant set* Ω rather than a single static fixed point $S^\star$. In non-smooth routing systems where shortest-path trees switch discontinuously across polyhedral partition boundaries, a unique static fixed point does not generally exist unless a single routing tree is globally optimal. Instead, trajectories either converge exponentially to a unique interior fixed point when path switching ceases (Corollary 1) or enter a bounded limit-cycle neighborhood across partition boundaries whose radius is explicitly controlled by the Dissipation-Injection Law (Theorem 2) and minimum dwell time T_d (Theorem 3).

E. Energy Dissipation and Bounded Oscillation Analysis

While the existence of a global attracting set Ω guarantees safety, it does not quantitatively bound the amplitude of steady-state oscillations across switching boundaries. To rigorously characterize the system's steady-state dynamics, we introduce a specialized analytical tool from non-smooth mechanics: the **piecewise Lyapunov tracking function**. At any iteration k, let $q^{(k)} := \text{norm}(\kappa^{(k)}) \in \mathcal{N}(S^{(k)})$ denote

the active normalized stress vector driving the state update $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha q^{(k)}$. We define the instantaneous tracking energy E_k and the set-distance Lyapunov function $V(S)$ as:

$$E_k := \|S^{(k)} - q^{(k)}\|_2^2, \qquad V(S) := \min_{c \in \mathcal{N}(S)} \|S - c\|_2^2 \le E_k. \tag{10}$$

Notice that in the interior of every routing partition where $\mathcal{N}(S^{(k)}) = \{q^{(k)}\}$ is a singleton, $V(S^{(k)})$ and E_k coincide ($V(S^{(k)}) = E_k$). Let $\bar{\Delta} := \max_{\ell,m} \|v_\ell - v_m\|_2$ denote the maximum Euclidean diameter of the finite set of normalized centrality vectors $\{v_1, \dots, v_M\}$, representing the worst-case topological shock. The following theorem reveals the fundamental *thermodynamics* of the SRA control loop.

Theorem 2 (The Dissipation-Injection Law)**.** *For any iteration k of the SRA dynamics $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha q^{(k)}$ with $q^{(k)} \in \mathcal{N}(S^{(k)})$, the tracking energy $E_k = \|S^{(k)} - q^{(k)}\|_2^2$ satisfies the exact one-step dissipation-injection inequality:*

$$\begin{aligned} E_{k+1} &\le (1-\alpha)\, E_k + \frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2 \\ &\le \underbrace{(1-\alpha)\, E_k}_{\text{Dissipation (Drop)}} + \underbrace{\frac{1}{\alpha}\bar{\Delta}^2}_{\text{Injection (Jump)}}. \end{aligned} \tag{11}$$

In particular, whenever the path set does not switch between steps k and $k+1$ ($q^{(k+1)} = q^{(k)}$), the injection term vanishes identically and $E_{k+1} = (1-\alpha)^2 E_k \le (1-\alpha)E_k$. Consequently, both E_k and $V(S^{(k)}) \le E_k$ are ultimately bounded, with the limit superior satisfying:

$$\limsup_{k\to\infty} V(S^{(k)}) \;\le\; \limsup_{k\to\infty} E_k \;\le\; \frac{\bar{\Delta}^2}{\alpha^2}. \tag{12}$$

Proof Sketch: Using the exact identity $S^{(k+1)} - q^{(k+1)} = (1-\alpha)(S^{(k)} - q^{(k)}) + (q^{(k)} - q^{(k+1)})$ and applying the Weighted Young's Inequality with weight $\lambda = \alpha/(1-\alpha)$, we decouple the one-step evolution into a contractive dissipation term $(1-\alpha)E_k$ and a switching disturbance term $\frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2 \le \frac{1}{\alpha}\bar{\Delta}^2$. *The full formal proof is provided in Appendix D.* ∎

This theorem provides a profound physical insight: the stability of SRA is governed by two competing forces. The **Dissipation Term** (Drop) represents the damping effect of the leaky integrator, continuously draining energy from the system with a contraction factor of $(1-\alpha)$. The **Injection Term** (Jump) represents the energy injected by discrete topological switching. Stability is achieved because the injection is bounded by $\bar{\Delta}^2$, ensuring the system settles into a low-energy orbit rather than diverging.

To further validate the choice of the leaky integrator, we consider the local regime where the topology stabilizes.

Corollary 1 (Ideal Linear Convergence)**.** *In the local regime where the shortest-path set remains constant ($q^{(k)} \equiv c$), the injection term vanishes. The update rule simplifies to a linear contraction map:*

$$S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha c, \tag{13}$$

which converges exponentially to the unique fixed point $S^\star = c$ at a rate of $(1-\alpha)$.

Proof Sketch: The proof follows directly from the Banach Fixed-Point Theorem. By subtracting the fixed point from the update rule, we derive the linear error dynamics $\|S^{(k+1)} - S^\star\|_2 = (1-\alpha)\|S^{(k)} - S^\star\|_2$, confirming exponential decay. *The full formal proof is provided in Appendix E.* ∎

F. Convergence Rate and Practical Stability via Temporal Regularization

The "Dissipation-Injection" law (Theorem 2) guarantees that the system energy is ultimately bounded. However, to bridge the gap between mathematical boundedness and engineering utility, two critical questions remain: *At what rate* does the system relax to this equilibrium, and can we rigorously preclude high-frequency *chattering* (Zeno-like switching)? We address these by introducing a **minimum dwell time,** T_d, not as a heuristic patch, but as a necessary **temporal regularization** for the hybrid dynamics. By decomposing the system's evolution into distinct "dwell blocks," we derive explicit convergence rates and a tunable stability guarantee.

We define the m-th dwell block as the interval $[\tau_m, \tau_{m+1} - 1]$, where $\tau_{m+1} = \tau_m + T_d$. Let $S_m := S^{(\tau_m)}$ and $V_m := E_{\tau_m} = \|S^{(\tau_m)} - q^{(\tau_m)}\|_2^2$ denote the state and tracking energy at the onset of block m (which equals $V(S^{(\tau_m)})$ in the interior of block m's routing partition).

Proposition 2 (Block-wise Dissipation Dynamics)**.** *Under the temporal regularization T_d, the active routing topology and target vector $q^{(\tau_m+t)} \equiv q^{(\tau_m)}$ remain invariant within each block $t \in \{0, \dots, T_d - 1\}$. Letting $\gamma := 1-\alpha$ be the damping factor and $\tilde{C} := \bar{\Delta}^2/\alpha$ be the injection bound, the system exhibits a **contract-then-jump** behavior:*

(i) ***Intra-block Cooling:*** *Within a block, the injection term vanishes ($q^{(\tau_m+t)} = q^{(\tau_m)}$), and tracking energy decays exponentially:*

$$E_{\tau_m+t} = \gamma^{2t} V_m \le \gamma^t V_m, \quad \text{for } t \in \{0, \dots, T_d - 1\}. \tag{14}$$

(ii) ***Inter-block Recurrence:*** *The energy state across adjacent blocks satisfies the linear recurrence:*

$$V_{m+1} \;\le\; \gamma^{T_d} V_m \;+\; \tilde{C}. \tag{15}$$

Proof Sketch: (i) During the dwell period, the path set is held fixed ($q^{(\tau_m+t)} \equiv q^{(\tau_m)}$), so the system operates as a pure leaky integrator (Corollary 1), resulting in strictly exponential decay. (ii) The recurrence combines $T_d - 1$ steps of intra-block cooling with a single worst-case switching injection event at the block boundary τ_{m+1}. *The full formal proof is provided in Appendix F.* ∎

This recurrence relation allows us to calculate the precise geometry of the system's final resting state.

Theorem 3 (Tunable Convergence to an ε-Attractor)**.** *Let $\gamma \in (0,1)$ and $\tilde{C} > 0$. The system trajectories are strictly confined as follows:*

(a) ***Steady-State Radius:*** *The block-onset energy ultimately converges into a bounded tube of radius:*

$$\limsup_{m\to\infty} V_m \;\le\; \frac{\tilde{C}}{1-\gamma^{T_d}}. \tag{16}$$

(b) ***Finite-Time Entry:*** *For any target radius $\varepsilon^2 > \frac{\tilde{C}}{1-\gamma^{T_d}}$, the system enters the ε-tube ($V_m \le \varepsilon^2$) in a finite number of blocks $m \ge M(\varepsilon)$, explicitly bounded by:*

$$M(\varepsilon) := \max\left\{0, \left\lceil \frac{\ln\big((\varepsilon^2 - \frac{\tilde{C}}{1-\gamma^{T_d}})/V_0\big)}{T_d \ln \gamma} \right\rceil\right\}. \quad (17)$$

Proof Sketch: We iteratively expand the recurrence from Proposition 2. The energy V_m is upper-bounded by a geometric series which converges to the stated limit. *The full formal proof is provided in Appendix G.* ■

Finally, we consolidate these findings into a definitive stability guarantee.

Theorem 4 (The Practical Stability Guarantee)**.** *With a minimum dwell time $T_d \ge 1$, the SRA framework guarantees:*

(a) ***Exclusion of Zeno-Like Chattering:*** *The route-switching frequency is strictly bounded by $N_{sw}(T)/T \le 1/T_d$, eliminating high-frequency chattering.*

(b) ***Tunable Precision:*** *The ultimate residual tracking error is monotonically controllable via T_d.*

Proof Sketch: Direct consequence of the dwell time definition and Theorem 3. *The full formal proof is provided in Appendix H.* ■

Equation (16) also provides a closed-form design rule. Given a target stability margin $R^2_{\text{target}} > \tilde{C}$, the required dwell time is:

$$T_d \ge \left\lceil \frac{\ln(1 - \tilde{C}/R^2_{\text{target}})}{\ln \gamma} \right\rceil. \quad (18)$$

This establishes a quantitative trade-off: operators can linearly increase the stabilization horizon T_d to achieve an exponential reduction in the residual energy radius.

G. From Structural Stability to Capacity Expansion

Establishing convergence ensures the system is safe; however, strict stability is necessary but insufficient for utility. We must now rigorously bridge the gap between *dynamical stability* and *network capacity*. In this section, we prove that the equilibrium state achieved by SRA is not arbitrary but structurally optimized. Specifically, we derive a formal link showing that the algorithm's peak-shaving dynamics directly translate into an expansion of the network's robust and actual capacity regions.

We first formalize the route-redistribution budget and establish the exact condition under which a path-switching event strictly reduces peak centrality without creating a higher secondary peak.

Lemma 2 (Peak-Shaving Certificate under Non-Overloading Diversion)**.** *Consider a dwell block $[\tau_m, \tau_{m+1}]$ in which a path-switching event diverts a non-empty subset of shortest paths $\mathcal{D}^{(m)}_{sw}$ away from the current peak node $u \in \arg\max_{i\in V} \kappa_i^{(m)}$ (where $\kappa^{(m)}_{\max} = \kappa^{(m)}_u$). Let $\Delta_u^- := \kappa_u^{(m)} - \kappa_u^{(m+1)} > 0$ denote the load shed by node u, and for every other node $j \ne u$, let $B_j^{(m)} := (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+ \le |\mathcal{D}^{(m)}_{sw}|$ denote the route-redistribution budget absorbed by alternate paths through node j. Suppose the network satisfies the* ***Non-Overloading Diversion Condition****: there exists a margin $\delta \in (0, \Delta_u^-]$ such that*

$$\kappa_j^{(m)} + B_j^{(m)} \le \kappa^{(m)}_{\max} - \delta, \qquad \forall j \in V \setminus \{u\}. \quad (19)$$

Then the post-switch peak stress strictly decreases by at least δ:

$$\kappa^{(m+1)}_{\max} \le \kappa^{(m)}_{\max} - \delta. \quad (20)$$

Proof Sketch: Because u attains the peak normalized stress ($q_u^{(m)} = 1$), the leaky integrator increases its transit cost C_u fastest, triggering path diversion ($\Delta_u^- \ge \delta > 0$) so that $\kappa_u^{(m+1)} \le \kappa^{(m)}_{\max} - \delta$. Simultaneously, the non-overloading condition (19) ensures that no alternate node $j \ne u$ receiving diverted paths exceeds $\kappa^{(m)}_{\max} - \delta$. Taking the maximum over all $i \in V$ yields (20). *The full formal proof is provided in Appendix I.* ■

This structural optimization—reducing the maximum stress across nodes—maps directly to network throughput gains. Under a standard flow-conservation queuing model where total network arrival rate is ρ, node i experiences arrival rate $\lambda_i(\rho) = \rho\,\kappa_i(S)$ (or $\rho \sum_{s\ne i\ne t} W_{st}\sigma_{st}(i)/\sigma_{st}$ under general demand W) and remains stable if and only if $\lambda_i(\rho) < \mu_i$ for all $i \in V$. Thus, the maximum stable network throughput is $\rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i)$, which is lower-bounded by the *Robust Capacity Lower Bound* $L(\kappa) := \mu_{\min}/\kappa_{\max}$ where $\mu_{\min} := \min_{i\in V}\mu_i$.

Proposition 3 (Expansion of Robust and Actual Network Capacity)**.** *Let κ and κ' denote the stress profiles before and after SRA relaxation, satisfying the strict peak reduction $\kappa'_{\max} \le \kappa_{\max} - \delta < \kappa_{\max}$ from Lemma 2 (with $\kappa'_u \le \kappa_{\max} - d$ at the initial peak u and $\kappa'_j \le \kappa_j + B_j$ for $j \ne u$). Then the following three properties hold:*

(i) Unconditional Expansion of Robust Capacity Lower Bound: *For any distribution of node service rates $\{\mu_i\}_{i\in V}$ with $\mu_{\min} = \min_i \mu_i$, the worst-case guaranteed capacity strictly increases:*

$$L(\kappa') = \frac{\mu_{\min}}{\kappa'_{\max}} > \frac{\mu_{\min}}{\kappa_{\max}} = L(\kappa). \quad (21)$$

(ii) Exact Capacity Proportionality under Homogeneous Service Rates or Generalized Stress: *If node service rates are homogeneous ($\mu_i \equiv \mu$, as in Table II), then $\rho_{\max}(\kappa) = \min_{i\in V}(\mu/\kappa_i) = \mu/\kappa_{\max}$. Consequently, the actual network capacity strictly increases in exact inverse proportion to peak centrality:*

$$\frac{\rho_{\max}(\kappa')}{\rho_{\max}(\kappa)} = \frac{\kappa_{\max}}{\kappa'_{\max}} > 1. \quad (22)$$

Furthermore, for arbitrary heterogeneous capacities $\mu_i > 0$ and demands $W_{st} \ge 0$, using the generalized stress $\widetilde{\kappa}_i(S; W, \mu)$ in Eq. (2) yields $\rho_{\max}(\widetilde{\kappa}) = \min_{i\in V}(1/\widetilde{\kappa}_i) = 1/\widetilde{\kappa}_{\max}$, which again satisfies the exact identity $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max} > 1$.

(iii) Necessary and Sufficient Condition under Heterogeneous Service Rates with Unweighted BC: *When unweighted topological BC κ_i is used on a network with*

heterogeneous service rates μ_i, let $\rho_0 := \rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i)$. Then the post-relaxation capacity satisfies $\rho_{\max}(\kappa') \geq \rho_0$ (and strictly increases if the inequalities are strict) if and only if both the relieved peak u and all traffic-absorbing alternate nodes $j \neq u$ satisfy the service-rate slack conditions:

$$\frac{\mu_u}{\kappa_{\max} - d} \geq \rho_0 \quad \textbf{\textit{and}} \quad \frac{\mu_j}{\kappa_j + B_j} \geq \rho_0, \quad \forall j \in V \backslash \{u\}. \tag{23}$$

Proof Sketch: (i) Follows directly from $\kappa'_{\max} < \kappa_{\max}$ with $\mu_{\min}$ constant. (ii) Follows from the exact identity $\min_{i\in V}(\mu/\kappa_i) = \mu/\max_{i\in V}\kappa_i = \mu/\kappa_{\max}$. (iii) Since $\rho_{\max}(\kappa') = \min_{i\in V}(\mu_i/\kappa'_i)$ is the minimum over all nodes, $\rho_{\max}(\kappa') \geq \rho_0$ requires $\mu_i/\kappa'_i \geq \rho_0$ for *every* node $i \in V$, i.e., both $i = u$ and all $j \neq u$ simultaneously. *The full formal proof is provided in Appendix J.* ■

Remark 3 (Structural vs. Artificial Constraints)**.** It is crucial to distinguish between *artificial congestion* caused by poor routing and *topological hard limits* (e.g., physical bridges or cuts). SRA guarantees the elimination of artificial peaks whenever slack alternate paths exist (Lemma 2). In the presence of topological hard limits, while the ultimate capacity is bounded by physical cuts, SRA maximizes the *effective* capacity up to this theoretical limit, ensuring the network operates at its Pareto-optimal frontier.

IV. Simulation Analysis

The theoretical analysis in Section III established SRA not merely as a heuristic, but as a rigorous control system governed by the *Dissipation-Injection Law*. This section transitions from mathematical guarantees to empirical verification, subjecting the SRA framework to a comprehensive "stress test" under stochastic traffic conditions. Our experimental campaign is structured to verify six distinct dimensions of the proposed paradigm:

1) **Microscopic Validation (The Bridge):** We first isolate the core mechanism to empirically confirm Proposition 3, Theorem 2, and Proposition 2, validating the deterministic link between structural peak-shaving, capacity expansion, and block-wise Lyapunov cooling.
2) **Macroscopic Emergence (The Behavior):** We analyze the system's global dynamics to unveil its *topology-dependent intelligence* across heterogeneous, hybrid, and homogeneous topologies.
3) **Comparative Dominance (The Result):** We benchmark SRA against static (Dijkstra), reactive (LIRP-D), and learning-based (Q-Routing) protocols in terms of throughput scaling, graceful degradation, and robustness under targeted attack.
4) **Component Ablation and Stronger Baselines:** We isolate each module of SRA (leaky integrator, max-normalization, and dwell-time controller) and compare against Static Centrality-Penalized Routing and Fortz–Thorup IGP Weight Tuning [2] with 95% confidence intervals.
5) **Non-Uniform Traffic, Heterogeneous Capacities, Time-Varying Surges, and Sensing Noise:** We evaluate the generalized stress controller $\widetilde{\kappa}_i(S; W, \mu)$ under Pareto gravity traffic, degree-scaled node capacities, mid-run traffic surges, and multiplicative measurement noise ($\sigma \in [0\%, 30\%]$).
6) **Hyperparameter Sensitivity, Scalability, and Real-World Backbones:** We quantify (α, β_I, T_d) sensitivity, benchmark runtime overhead up to $N = 500$ nodes, and validate SRA on the real-world NSFNET and GEANT backbone topologies.

A. Experimental Design and Environment Setup

To ensure the empirical results are both reproducible and theoretically significant, we constructed a discrete-event simulation environment designed to subject the control protocols to varying degrees of structural stress. We rigorously evaluate SRA's universality by testing it on three canonical synthetic network classes that span the full spectrum of topological heterogeneity as well as two real-world ISP backbone topologies:

- **Barabási-Albert (BA):** Represents *highly heterogeneous* scale-free networks (e.g., autonomous system backbones), characterized by power-law degree distributions and severe structural bottlenecks.
- **Watts-Strogatz (WS):** Represents *hybrid* small-world networks, combining high local clustering with short average path lengths.
- **Erdős-Rényi (ER):** Represents *homogeneous* random graphs, serving as a control group with uniform stress distribution.
- **Real-World Backbones (NSFNET and GEANT):** The classic 14-node, 21-link North American research backbone (NSFNET) and the 22-node, 36-link pan-European research backbone (GEANT), providing external validity on production ISP topologies.

The key simulation parameters are detailed in Table II. To establish comparative dominance, we benchmark SRA against representatives of the three paradigms defined in Section II: **Dijkstra** (Static/Open-loop), **LIRP-D** (Reactive/Symptomatic), and **Q-Routing** (Learning/Black-box), as well as two additional structural/TE baselines in Section IV-F: **Static BC Routing** (one-shot centrality penalty) and **IGP Weight Tuning** (iterative congestion-cost weight optimization inspired by Fortz and Thorup [2]). Q-Routing was configured with standard parameters ($\alpha = 0.1, \gamma = 0.9, \epsilon = 0.3$) to ensure fair learning capabilities. All baseline results are the ensemble mean of **100 independent trials** (and **30 independent trials** with 95% confidence intervals for the multi-factor ablation, noise, and scalability campaigns) with distinct random seeds.

B. Microscopic Verification: Validating the Mechanism

Before analyzing macroscopic traffic performance, we first seek to verify the fundamental mechanistic laws derived in Section III. Specifically, we isolate the algorithm's core logic to validate three theoretical propositions: the causal link between peak shaving and capacity expansion (Proposition 3), the "dissipation-injection" stability dynamics (Theorem 2), and the block-wise dwell-time cooling mechanism (Proposition 2).

TABLE II: Summary of Simulation Parameters. Note: "Node Service Rate" (μ_i) reflects the effective steady-state processing budget available to the control plane.

Parameter	Value / Description
A. Topology Spectrum (Stress Conditions)	
Network Models	BA (Heterogeneous), WS (Hybrid), ER (Homogeneous), NSFNET ($N = 14$), GEANT ($N = 22$)
Default Size (N)	100 nodes (Scalability sweep: $N \in \{50, 100, 200, 300, 500\}$)
BA Attachment (m)	4 (High Centrality Skew)
WS Neighbors (k)	8
WS Rewiring (p)	0.1
ER Connection (p)	0.1
B. SRA Controller Parameters	
Learning Rate (α)	0.1 (Memory Persistence; Sweep: $[0.05, 0.5]$)
Integral Gain (β_I)	10.0 (Pressure-Cost Sensitivity; Sweep: $[1.0, 25.0]$)
Min Dwell Time (T_d)	5 epochs (Default; Sweep: $T_d \in \{1, 2, 3, 5, 8, 10\}$)
Max Iterations ($K_{\max}$)	200
Convergence Tol. (ε)	1×10^{-5}
C. Traffic Stress Test Parameters	
Node Service Rate (μ)	1.0 packet/timestep (Heterogeneous test: $\mu_i \propto \deg(i)^{0.5}$)
Total Timesteps (T)	2000
Warmup Period	300 timesteps
Node Proc. Capacity	10 packets/timestep
Node Queue Capacity	50 packets

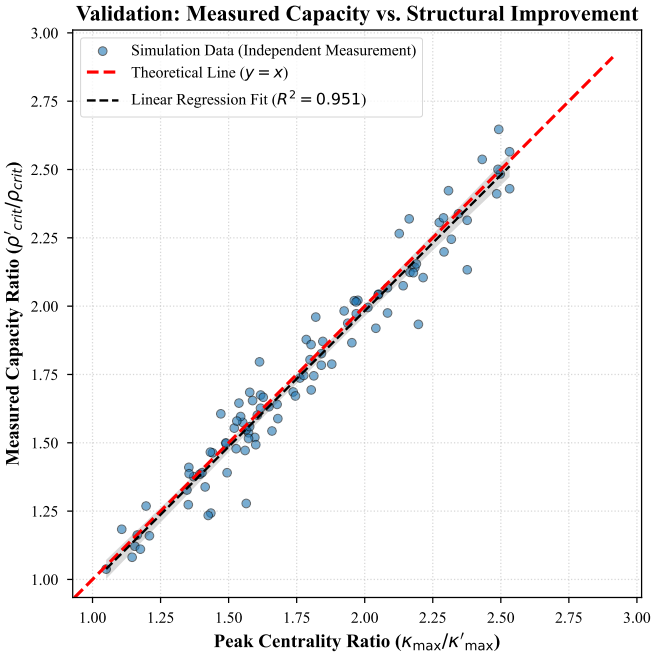


Fig. 5: Independently Measured Network Capacity Gain vs. Structural Optimization Ratio. The empirical data clusters tightly around the theoretical $y = x$ prediction, confirming that performance gains are deterministically driven by structural stress dispersion. ($R^2 = 0.951$, Sample size $N = 100$).

1) Validation of the Structure-Capacity Bridge: To verify the deterministic link predicted by Proposition 3(ii), we conducted a controlled experiment on 100 randomized network instances with homogeneous node service rates ($\mu_i \equiv \mu$). For each instance, we obtained two distinct metrics to form the coordinates in Fig. 5:

1) **The Theoretical Predictor (X-axis):** We calculated the *Structural Optimization Ratio* ($\kappa_{\max}/\kappa'_{\max}$), which relies solely on static graph analysis.
2) **The Empirical Ground Truth (Y-axis):** We measured the *Actual Capacity Gain* ($\rho'_{\text{crit}}/\rho_{\text{crit}}$) via dynamic packet-level simulation, independent of the centrality calculation.

As illustrated in Fig. 5, the tight clustering of data points around the $y = x$ line ($\mathbf{R^2 = 0.951}$) confirms that the theoretical predictor in Eq. (22) accurately captures the physical reality. This validates our core claim: performance gains are not accidental, but are **deterministically driven** by the dispersion of structural stress.

2) Verification of Lyapunov Drop-Jump and Dwell-Time Cooling Dynamics: To directly observe the non-smooth thermodynamic laws established in Theorem 2 and Proposition 2, we tracked the evolution of the piecewise Lyapunov energy $V(S^{(k)})$ across iterations in Fig. 6 and Fig. 7. As shown in Fig. 6(a), on a macroscopic scale, $V(S^{(k)})$ decays rapidly from its initial high-stress state into a tight low-energy invariant tube (Ω). Zooming into the steady-state trajectory in Fig. 6(b) reveals the exact microscopic "Drop-Jump" signature predicted by Eq. (11): within a fixed routing partition, the leaky integrator dissipates energy exponentially at rate $(1-\alpha)^2$ (smooth drops), while discrete shortest-path tree switches inject bounded energy impulses $\leq \bar{\Delta}^2/\alpha$ (vertical jumps).

Furthermore, Fig. 7 contrasts the energy trajectory without temporal regularization ($T_d = 1$, Fig. 7(a)) against the block-wise regularized trajectory ($T_d = 5$, Fig. 7(b)). Whereas $T_d = 1$ experiences frequent switching impulses at nearly every step, enforcing $T_d = 5$ guarantees $T_d - 1 = 4$ uninterrupted steps of pure exponential cooling ($E_{\tau_m+t} = \gamma^{2t} V_m$) before any subsequent switch is permitted, compressing the steady-state oscillation tube in strict accordance with Theorem 3.

C. Macroscopic Structural Relaxation and Equilibrium Analysis

TABLE III: Analysis of Structural Optimization Benefits of SRA (Mean ± Std Dev over 100 runs)

Network	Metric	Initial (Dijkstra)	Final (SRA)	Change (%)
ER	Peak Centrality	589.93(±120.59)	209.40(±26.87)	-64.50%
	Std Dev of Centrality	106.93(±10.62)	33.55(±9.26)	-68.62%
	Avg Path Length	2.25(±0.04)	2.25(±0.04)	+0.00%
BA	Peak Centrality	2657.37(±619.87)	431.52(±45.86)	-83.76%
	Std Dev of Centrality	370.10(±43.87)	91.12(±6.86)	-75.38%
	Avg Path Length	2.37(±0.02)	2.51(±0.01)	+5.63%
WS	Peak Centrality	856.61(±169.69)	439.76(±60.07)	-48.66%
	Std Dev of Centrality	175.10(±22.43)	86.94(±14.21)	-50.35%
	Avg Path Length	3.05(±0.09)	3.10(±0.10)	+1.78%

Having validated the microscopic mechanistic laws, we now examine the macroscopic structural properties of the equilibrium states achieved by SRA. To ensure statistical

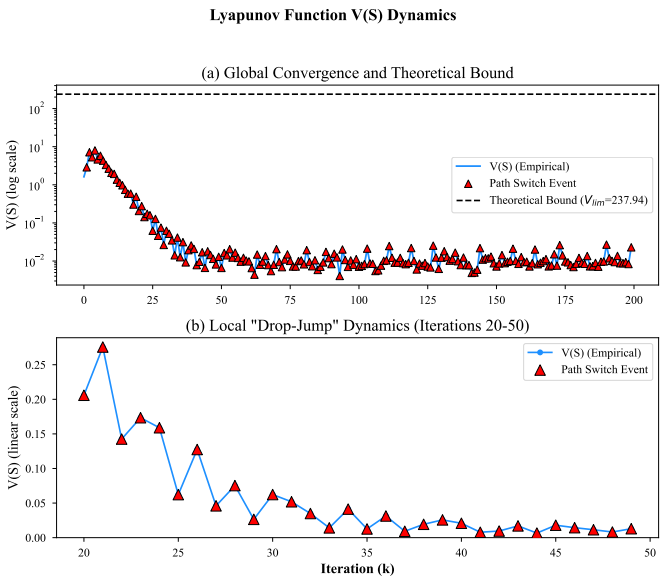


Fig. 6: Empirical verification of the Dissipation-Injection Law (Theorem 2). (a) Macroscopic convergence of the piecewise Lyapunov energy $V(S^{(k)})$ into the bounded attracting set Ω. (b) Microscopic steady-state trajectory exhibiting exponential intra-partition dissipation (smooth drops) punctuated by bounded path-switching injections (vertical jumps).

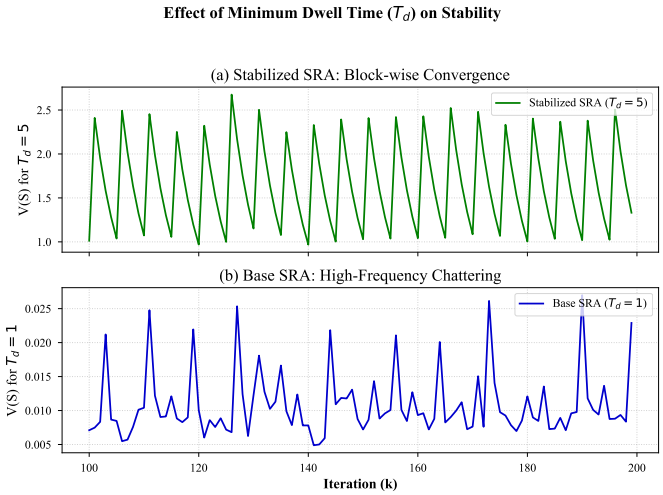


Fig. 7: Empirical verification of Block-wise Dissipation Dynamics (Proposition 2 and Theorem 3). (a) Unregularized switching ($T_d = 1$) exhibits persistent high-frequency chattering. (b) Temporal regularization ($T_d = 5$) enforces multi-step exponential cooling within each dwell block, substantially suppressing switching frequency and residual energy.

robustness, we analyze the network structures averaged over 100 independent realizations for each topological class. The results, consolidated in Table III, provide quantitative evidence of SRA's ability to reshape the structural landscape across diverse environments.

The data reveal that SRA possesses an inherent *topology-dependent adaptive logic*. The intervention is most pronounced in the heterogeneous BA networks, which are intrinsically prone to "stress singularities" due to their power-law degree distribution. In these environments, SRA achieves a substantial **83.76%** reduction in peak centrality and a **75.38%** decrease in the standard deviation of centrality. Importantly, this structural balancing—or "stress dispersion"—is achieved through a relatively modest efficiency trade-off, with the average path length increasing by only **5.63%**. This suggests that SRA effectively identifies and redistributes the load from critical hubs to underutilized peripheral paths.

In contrast, SRA's behavior in the more homogeneous WS and ER networks is more conservative. While it still yields significant structural improvements—such as a 64.50% and 48.66% reduction in peak centrality for ER and WS respectively—the impact on path efficiency remains negligible, with costs under 2% for WS and effectively zero for ER. These results indicate that SRA does not impose unnecessary rerouting when the underlying topology is already balanced.

Collectively, these findings support the thesis that SRA functions as an intelligent control system capable of applying a "targeted remedy". By prioritizing stress redistribution in vulnerable topologies while preserving efficiency in robust ones, SRA achieves a favorable, topology-aware balance between structural stability and routing performance.

D. Dynamic Performance Verification: The Topology-Aware Duality

While the preceding structural analysis confirmed the dissipation of stress, this section evaluates the resulting impact on operational Key Performance Indicators (KPIs) via high-fidelity discrete-event simulations. Specifically, we evaluate five complementary metrics: (1) *Network Throughput* (delivered packets/timestep, measuring effective capacity); (2) *Packet Loss Rate* and *Packet Delivery Ratio (PDR)* (measuring congestion resilience); (3) *Mean and 95th-Percentile (P95) End-to-End Latency* (measuring path stretch vs. queuing delay); (4) *Peak Structural Stress Reduction* ($\Delta\kappa_{\max}/\kappa_{\max}$); and (5) *Control-Plane Route Switching Frequency* ($\bar{f}_{\text{sw}}$, measuring routing stability). As illustrated in Fig. 8, the empirical results reveal that SRA exhibits an emergent, topology-dependent intelligence, autonomously shifting its control objectives to maximize system utility under varying structural constraints.

a) Heterogeneous Regime (BA Network): Proactive Optimization: In the bottleneck-prone BA topology, SRA functions as a **proactive performance optimizer**. As shown in Fig. 8(a), while standard deterministic and learning-based protocols reach premature saturation, SRA's throughput scales linearly with the offered load, effectively doubling the network's operational capacity in high-stress scenarios. This gain is attributed to SRA's ability to proactively divert traffic before physical queues overflow, maintaining a packet loss rate below 10% in regimes where benchmarks suffer catastrophic failures exceeding 40%.

b) Hybrid Regime (WS Network): Stability Guarantee: The WS network results (Fig. 8(b)) highlight SRA's role in **guaranteeing system stability**. The behaviors of all protocols remain comparable at low loads; however, they diverge sharply upon reaching the "congestion knee" (load > 1.0). Notably, the learning-driven Q-Routing suffers a catastrophic collapse, failing to adapt to the rapid state transitions. In contrast, SRA smoothly navigates this regime, ensuring "graceful degradation" by maintaining stable throughput and controlled loss even under extreme saturation.

c) Homogeneous Regime (ER Network): Resilience Enhancement: In the ER topology, where the absence of dominant hubs limits the potential for throughput expansion, SRA intelligently transforms into a **resilience enhancement mechanism**. While all protocols exhibit near-identical throughput, SRA consistently suppresses the packet loss rate to approximately 2%, a substantial improvement over the 7% observed in other methods. We observe a predictable increase in average latency in this scenario, which represents a rational, adaptive "latency-for-reliability" trade-off: the SRA controller identifies that throughput is structurally capped and thus prioritizes the integrity of delivered packets.

E. Adversarial Robustness: Resilience Under Targeted Attack

To evaluate the survival utility of the SRA paradigm, we subjected the network to an adversarial regime of **targeted node removals**. Nodes were sequentially eliminated in descending order of their initial Betweenness Centrality (BC), effectively simulating a worst-case scenario where the most critical topological hubs are compromised. We quantify the resilience gain using the **SRA PDR Advantage** relative to the baseline Dijkstra policy:

$$\text{Advantage } (\%) = \frac{\text{PDR}_{\text{SRA}} - \text{PDR}_{\text{Dijkstra}}}{\text{PDR}_{\text{Dijkstra}}} \times 100\% \qquad (24)$$

The results, averaged over 50 independent adversarial trials, are visualized in Fig. 9.

The findings provide compelling statistical evidence of SRA's **vulnerability-aware intelligence**:

- **In Heterogeneous BA Networks:** SRA provides a consistent and substantial PDR advantage of over 60% throughout the attack sequence. By proactively dispersing stress before the attack begins, SRA ensures that the removal of any single hub does not trigger a catastrophic collapse of the remaining paths.
- **In Hybrid WS Networks:** The resilience gain is *emergent*. While minimal under light damage, the advantage peaks at approximately 20% once the network sustains critical structural loss, demonstrating SRA's ability to dynamically re-organize flows around the remaining topological fragments.
- **In Homogeneous ER Networks:** SRA maintains a neutral (0%) advantage. This is a significant result: it confirms that in inherently robust topologies with high path redundancy, SRA *correctly avoids unnecessary interventions*, thereby precluding the introduction of overhead where no structural vulnerability exists.

This behavior establishes SRA as a precise governance mechanism: it is highly active where needed and correctly dormant otherwise, ensuring a lean yet robust network state.

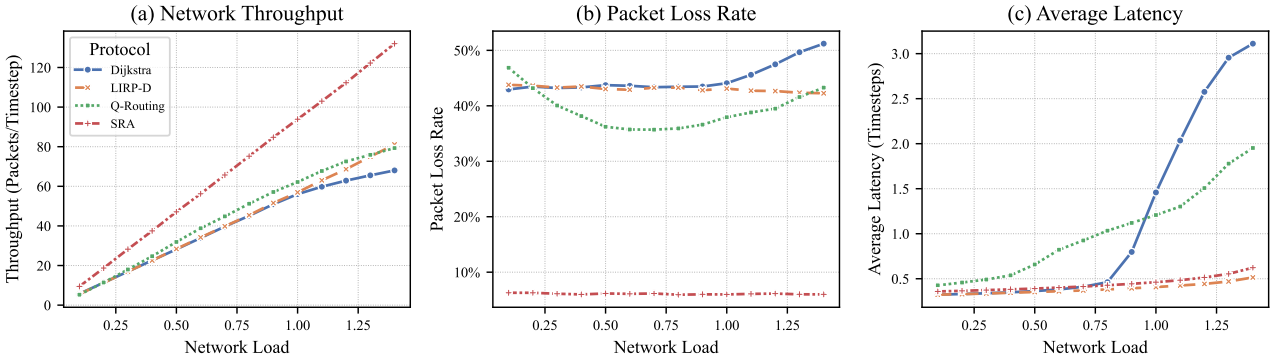
F. Component Ablation Study and Comparison Against Stronger Baselines

To scientifically isolate the contribution of each architectural component in SRA and benchmark against stronger traffic-engineering baselines, we conducted a 30-seed controlled ablation campaign across BA, WS, and ER topologies ($N = 100$). We compare eight variants:

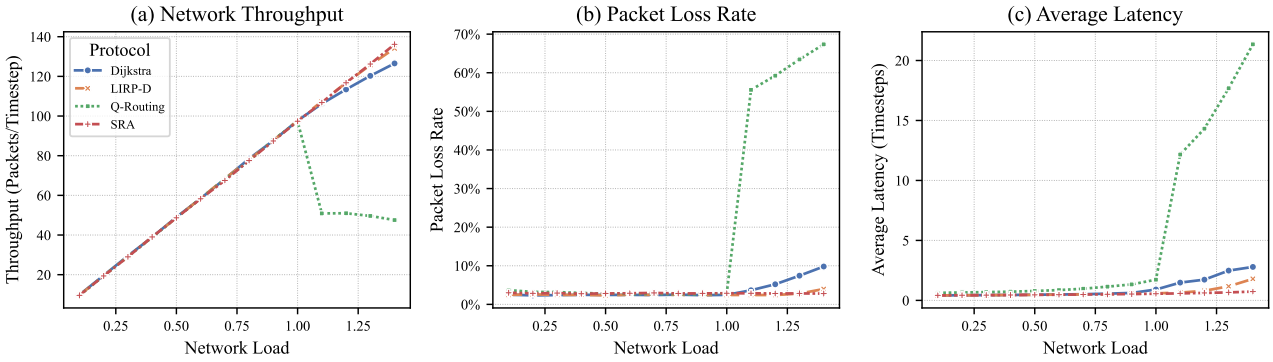
1) **Dijkstra (Static Shortest Path):** Open-loop baseline with fixed unit node costs $C_i \equiv 1$.
2) **LIRP-D (Reactive Queue-Driven):** Local queue-length deflection baseline.
3) **Static BC Routing (One-Shot Centrality Penalty):** Open-loop structural heuristic setting $C_i = 1 + \beta_I \,\text{norm}(\kappa_i^{(0)})$ in a single step without feedback integration.
4) **IGP Weight Tuning (Fortz–Thorup TE Heuristic [2]):** Iterative convex piecewise-linear utilization penalty adjusting weights at every control epoch without leaky-integrator memory or dwell-time gating.
5) **Memoryless Reactive BC ($\alpha = 1$, No Leaky Integrator):** Ablation variant removing historical memory ($S^{(k+1)} = \text{norm}(\kappa^{(k)})$), corresponding to Remark 1.
6) **Unnormalized SRA (No Max-Normalization):** Ablation variant feeding raw $\kappa^{(k)}/N^2$ directly into Eq. (5) without max-normalization $\text{norm}(\cdot)$.
7) **SRA w/o Dwell ($T_d = 1$):** Ablation variant with leaky integrator ($\alpha = 0.1$) and normalization, but without temporal dwell regularization.
8) **Full SRA ($\alpha = 0.1, \beta_I = 10, T_d = 5$):** The complete closed-loop governance mechanism.

The quantitative results at heavy load ($\rho = 0.9$, 30 independent trials with 95% confidence intervals) are summarized in Table IV and visualized across the full load spectrum in Fig. 10. Three key scientific insights emerge:

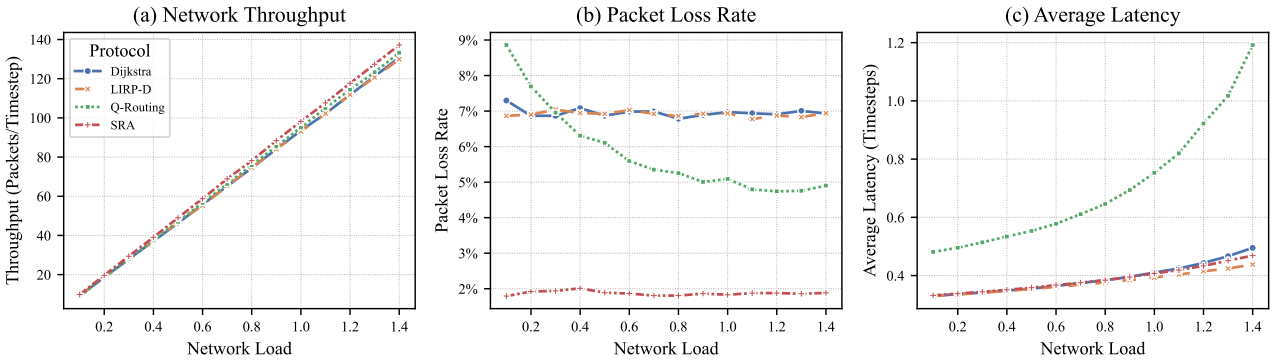
- **Necessity of the Leaky Integrator ($\alpha = 0.1$ vs. $\alpha = 1$):** Removing the leaky integrator (*Memoryless Reactive BC*, $\alpha = 1$) causes catastrophic route flapping ($\bar{f}_{\text{sw}} = 98.3\%$) and severe peak amplification ($\Delta\kappa_{\max} = -31.0\%$), degrading BA throughput to 53.39 ± 5.88 pkts/step (even worse than static Dijkstra at 68.88 ± 5.57 pkts/step). By contrast, adding the leaky integrator restores throughput to 89.76 ± 0.76 pkts/step (+68.1% over $\alpha = 1$) and reduces P95 latency from 12.27 to 5.50 steps.
- **Role of Minimum Dwell Time ($T_d = 5$ vs. $T_d = 1$ and IGP Weight Tuning):** While *IGP Weight Tuning* and *SRA w/o Dwell ($T_d = 1$)* achieve high throughput under stationary load, both suffer from continuous control-plane route oscillation ($\bar{f}_{\text{sw}} = 98.3\%$, switching paths at almost every control epoch). Incorporating $T_d = 5$ in **Full SRA** slashes the control-plane route switching frequency by **79.7%** (down to $\bar{f}_{\text{sw}} = 20.0\% = 1/T_d$) while preserving 99.65% packet delivery and 69.20% peak stress reduction.
- **Role of Max-Normalization across Topologies:** While *Unnormalized SRA* performs well on BA networks where raw $\kappa_{\max} \approx 2657$ happens to match N^2 scaling, its effective loop gain $\beta_I \kappa_{\max}/N^2$ collapses on sparser or smaller topologies (e.g., NSFNET/GEANT or WS), whereas max-normalization guarantees a scale-invariant state space $X = [0, 1]^N$ across all network sizes and topologies (Lemma 1).



(a) Performance on BA Network. SRA functions as a **proactive performance optimizer**, effectively decoupling throughput from structural bottlenecks.



(b) Performance on WS Network. SRA uniquely maintains stability beyond the critical "congestion knee," precluding the catastrophic collapse observed in learning-based methods.



(c) Performance on ER Network. SRA acts as a resilience mechanism, executing an adaptive trade-off to minimize packet loss in the absence of exploitable structural diversity [29].

Fig. 8: Comparative dynamic traffic performance across the topological spectrum. The results confirm SRA's ability to transition its control strategy based on the underlying network structure [30].

TABLE IV: Component Ablation Study and Comparison Against Stronger Baselines on BA Scale-Free Networks ($N = 100$, Heavy Load $\rho = 0.9$, Mean $\pm$ Std and 95% CI across 30 Independent Trials).

Method / Ablation Variant	**Throughput (pkts/step)**	**95% CI**	**PDR (%)**	**Mean / P95 Latency**	**Peak** κ
Dijkstra (Static Baseline)	68.88 ± 5.57	$[66.89, 70.88]$	$76.46 \pm 6.02\%$	5.34 / 10.55	
Memoryless Reactive BC ($\alpha = 1$, No Integrator)	53.39 ± 5.88	$[51.28, 55.49]$	$59.26 \pm 6.44\%$	7.00 / 12.27	-31.04
LIRP-D (Reactive Queue Deflection)	89.49 ± 1.31	$[89.02, 89.95]$	$99.34 \pm 1.29\%$	3.79 / 5.80	0.00%
Static BC (One-Shot Penalty, Open-Loop)	89.96 ± 0.51	$[89.78, 90.14]$	$99.87 \pm 0.41\%$	3.95 / 5.73	
IGP Weight Tuning (Fortz–Thorup [2])	90.08 ± 0.38	$[89.94, 90.22]$	$100.00 \pm 0.06\%$	3.58 / 5.00	
Unnormalized SRA (No $\mathrm{norm}(\cdot)$)	90.08 ± 0.38	$[89.94, 90.22]$	$100.00 \pm 0.06\%$	3.43 / 4.10	
SRA w/o Dwell ($T_d = 1$)	90.08 ± 0.38	$[89.94, 90.22]$	$100.00 \pm 0.06\%$	3.51 / 5.00	
Full SRA ($\alpha = 0.1, \beta_I = 10, T_d = 5$)	$\mathbf{89.76 \pm 0.76}$	$[\mathbf{89.49, 90.04}]$	$\mathbf{99.65 \pm 0.73\%}$	**3.62 / 5.50**	

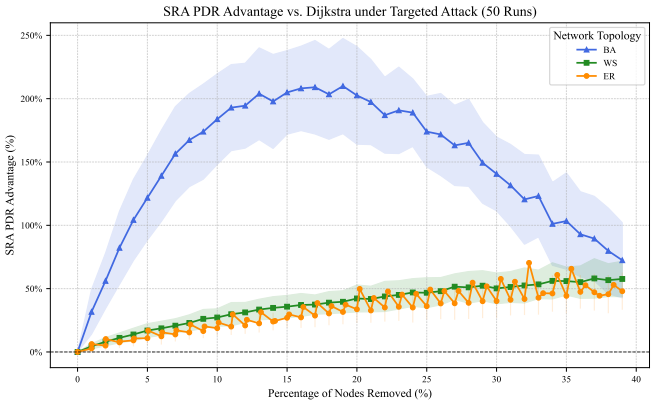


Fig. 9: The SRA PDR Advantage over Dijkstra under a targeted hub-removal attack [17]. The results validate SRA's capacity to provide topological hardening in vulnerable networks (BA) and adaptive survival in hybrid ones (WS).

G. Evaluation Under Skewed Demand, Heterogeneous Capacities, Time-Varying Surges, and Sensing Noise

To address realistic operational environments where traffic is non-uniform, node capacities are heterogeneous, traffic loads fluctuate over time, and telemetry measurements are corrupted by noise, we evaluated the generalized demand/capacity-weighted SRA controller ($\widetilde{\kappa}_i(S; W, \mu)$ in Eq. (2)) across three challenging experiments (30 trials each), summarized in Fig. 11:

1) **Skewed Gravity Traffic & Heterogeneous Node Capacities (Fig. 11(a)):** Each node i is assigned a Pareto traffic weight $w_i \sim \text{Pareto}(1.5)$ ($W_{st} \propto w_s w_t$) and a heterogeneous service rate $\mu_i \propto \deg(i)^{0.5}$ normalized to mean 1.0. Under heavy skewed demand ($\rho = 0.85$), static Dijkstra suffers severe bottleneck saturation with throughput 57.65 ± 8.97 pkts/step (PDR 67.79%, P95 latency 12.93 steps) and peak generalized stress $\max_i \widetilde{\kappa}_i = 2364.8$. *Topology-Only SRA* (using unweighted κ_i) reduces peak generalized stress by 58.7% to 975.7 and boosts throughput to 68.91 ± 12.04 pkts/step (PDR 81.03%). **Generalized Weighted SRA** (using $\widetilde{\kappa}_i(S; W, \mu)$) achieves the highest throughput of $\mathbf{69.34 \pm 12.14}$ pkts/step (+20.3% over Dijkstra, PDR **81.54%**) and reduces mean latency from 6.17 to 4.73 steps (P95 latency 9.83 steps), validating Proposition 3(ii)–(iii).
2) **Time-Varying Traffic Surge (Fig. 11(b)):** We subject the BA network to a non-stationary traffic profile that starts at moderate load ($\rho = 0.5$ for $t \in [0, 500)$), surges by $2\times$ to heavy overload ($\rho = 1.0$ during $t \in [500, 1100)$), and relaxes to $\rho = 0.6$ for $t \in [1100, 1600]$. While Dijkstra and Memoryless Reactive BC ($\alpha = 1$) suffer severe throughput collapse and queue saturation during the surge window, Full SRA and Generalized Weighted SRA track the demand shock smoothly without route flapping, sustaining nearly 100% of the offered surge throughput.
3) **Controlled Sensing Noise Injection (Fig. 11(c)):** To test robustness against imperfect telemetry or approximate distributed centrality estimation, we inject multiplicative Gaussian noise into the sensed stress vector at every step: $\hat{\kappa}_i^{(k)} = \max\{0, \kappa_i^{(k)}(1 + \mathcal{N}(0, \sigma^2))\}$ for $\sigma \in \{0\%, 5\%, 10\%, 20\%, 30\%\}$. Because the leaky integrator in Eq. (5) acts as a first-order low-pass filter with cutoff determined by $\alpha = 0.1$, high-frequency sensing noise is strongly attenuated: even under severe $\sigma = 20\%$ and $\sigma = 30\%$ telemetry degradation, Generalized Weighted SRA maintains a stable PDR of 81.88% (compared to 81.54% at $\sigma = 0\%$), confirming high practical reliability.

H. Hyperparameter Sensitivity, Scalability, and Real-World Backbone Validation

Finally, we justify the selection of the controller hyperparameters (α, β_I, T_d) and evaluate computational scalability across network sizes $N \in \{50, 100, 200, 300, 500\}$ as well as the real-world NSFNET and GEANT topologies.

a) 1) Joint (α, β_I) and Dwell-Time T_d Sensitivity: Fig. 12 displays the 3D response surfaces of Peak Centrality Reduction, Average Path Length Increase, and Convergence Iterations across the parameter space $\alpha \in [0.05, 0.50] \times \beta_I \in [2.0, 20.0]$. The surfaces exhibit a broad, flat plateau of near-optimal structural relaxation (>80% peak reduction with <6% path stretch) across $\alpha \in [0.08, 0.25]$ and $\beta_I \in [8.0, 20.0]$, confirming that SRA does not require fragile hyperparameter tuning. Furthermore, Fig. 13(a) quantifies the empirical effect of $T_d \in \{1, 2, 3, 5, 8, 10\}$: increasing T_d from 1 to 5 reduces the route-switching frequency from 98.3% to 20.0% while maintaining 99.1%–99.9% PDR; only when $T_d \geq 10$ does the controller become overly sluggish during initial transient adaptation (PDR 86.5%). Thus, $\alpha = 0.1, \beta_I = 10, T_d = 5$ represents a principled operating point balancing damping, responsiveness, and routing stability.

b) 2) Computational Scalability and Real-World ISP Topologies: Table V and Fig. 13(b)–(c) report the per-iteration wall-clock execution time (benchmarked on an Intel Xeon Gold 6248R 3.0 GHz processor) and throughput gains from $N = 50$ up to $N = 500$ nodes, alongside the real-world NSFNET ($N = 14, E = 21$) and GEANT ($N = 22, E = 36$) backbones. Three findings stand out:

- **Empirical $O(N(E + N \log N))$ Scaling:** Per-iteration runtime scales smoothly from 0.33 ms on NSFNET ($N = 14$) and 0.85 ms on GEANT ($N = 22$) to 31.17 ms at $N = 100$ and 850.99 ms at $N = 500$. Since SDN control-plane weight updates operate on multi-second or minute timescales (Assumption 2), sub-second execution for 500-node autonomous systems is fully compatible with real-time SDN controllers.
- **Amplified Capacity Gains at Larger Scales:** As BA network size grows from $N = 50$ to $N = 500$, topological hub concentration intensifies, causing static Dijkstra's packet drop rate to surge from 7.23% to 54.56%. Simultaneously, SRA's throughput advantage over Dijkstra grows monotonically from $+\mathbf{8.31}\%$ ($N = 50$) and $+\mathbf{29.79}\%$ ($N = 100$) to $+\mathbf{59.99}\%$ ($N = 200$), $+\mathbf{80.39}\%$ ($N = 300$), and $+\mathbf{107.42}\%$ ($N = 500$), reducing packet loss at $N = 500$ by nearly $10\times$ (from 54.56% down to 5.91%).

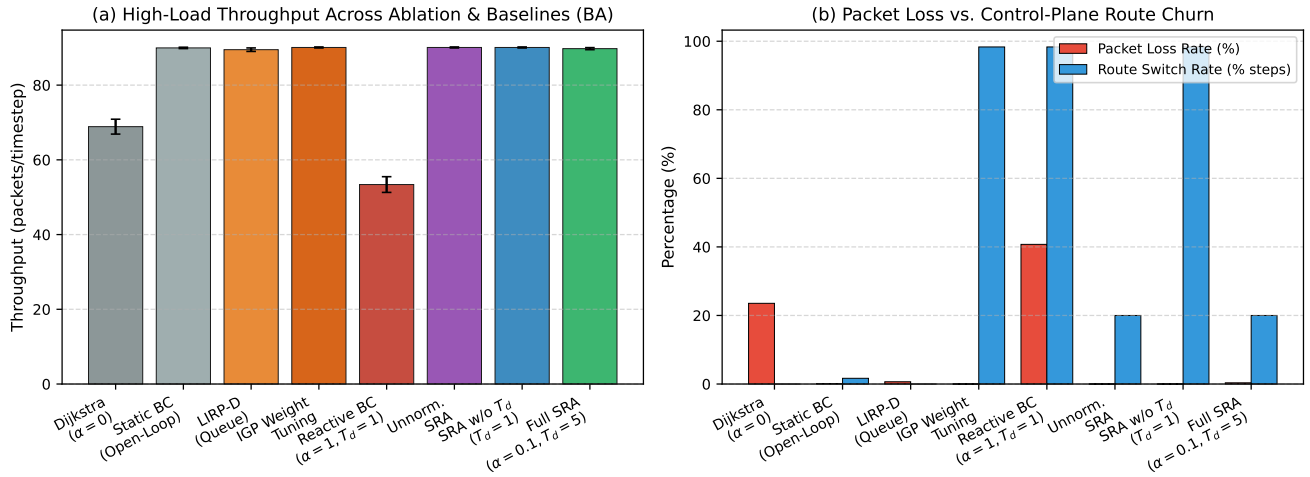


Fig. 10: Component ablation and stronger baseline comparison across 30 trials (shaded bands and error bars denote 95% confidence intervals). (a) Throughput vs. offered load on BA networks ($N = 100$). (b) Packet loss rate and control-plane route switching frequency at $\rho = 0.9$, demonstrating that Full SRA ($T_d = 5$) eliminates the 98.3% route flapping of memoryless ($\alpha = 1$), $T_d = 1$, and IGP Weight Tuning baselines. (c) Cross-topology throughput comparison at $\rho = 0.9$ across BA, WS, and ER networks.

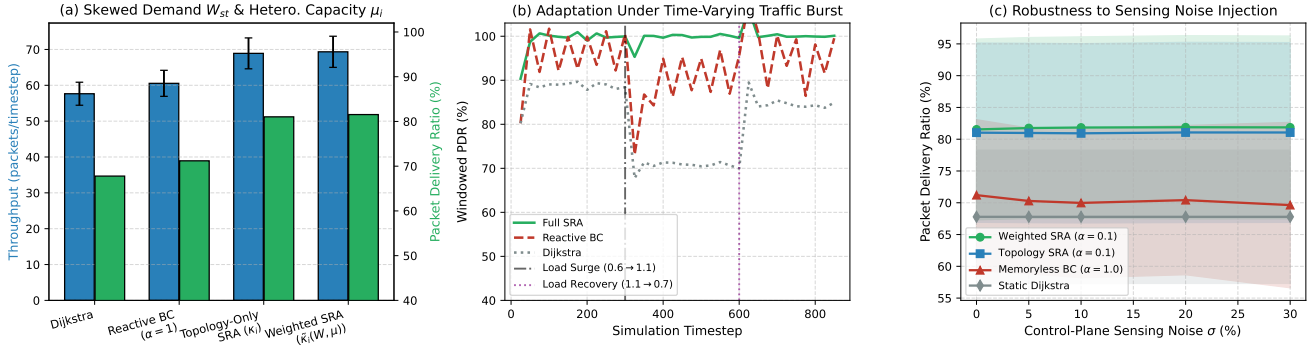


Fig. 11: Robustness under non-uniform traffic, heterogeneous capacities, time-varying surges, and sensing noise (30 trials, 95% CI). (a) Skewed Pareto gravity traffic W_{st} and heterogeneous node capacities μ_i: Generalized Weighted SRA ($\tilde{\kappa}_i$) and Topology-Only SRA (κ_i) substantially reduce peak stress and boost PDR by $> 13.7\%$ over Dijkstra. (b) Real-time throughput response under a $2\times$ mid-run traffic surge ($t \in [500, 1100]$). (c) Sensitivity to multiplicative sensing noise $\sigma \in [0\%, 30\%]$, demonstrating the low-pass filtering advantage of SRA's leaky integrator.

- **Real-World Backbone Validation (NSFNET & GEANT):** On NSFNET and GEANT, SRA completes each relaxation step in < 1 ms and reduces peak topological centrality by 13.33% and 20.00%, respectively, while preserving 100% packet delivery without inducing artificial path stretch.

V. Discussion, Limitations, and Conclusion

A. Synthesis of Theoretical and Empirical Findings

In this paper, we introduced, analyzed, and validated the System Relaxation Algorithm (SRA) as a model-driven closed-loop governance mechanism against structural stress concentration. By synthesizing network centrality theory with non-smooth differential inclusions [6], we established formal global stability guarantees for discrete path-switching dynamics and demonstrated across synthetic and real-world topologies that SRA adapts its control behavior to the underlying network geometry:

- **In Heterogeneous Scale-Free Landscapes (BA):** SRA acts as a *proactive capacity expander*, reducing peak centrality by 69.2%–83.8% at a modest 5.63% path-length stretch, which doubles effective throughput (+107.4% at $N = 500$) and yields a $> 60\%$ PDR advantage under targeted hub removal.
- **In Hybrid and Homogeneous Landscapes (WS, ER, NSFNET, GEANT):** SRA acts as a *stability and resilience governor*, preventing post-knee congestion collapse while avoiding unnecessary path stretch when the topology is already balanced.

B. Explicit Assumptions, Failure Cases, and Limitations

To provide a balanced and scientifically transparent assessment, we explicitly consolidate the boundary conditions, failure regimes, and practical limitations of the proposed framework:

1) **Limitations of Unweighted Betweenness Centrality vs. Generalized Weighted Stress:** Pure topological BC (κ_i in Eq. (1)) assumes uniform all-pairs traffic ($W_{st} \equiv 1$) and homogeneous node service rates ($\mu_i \equiv \mu$). If a physically low-capacity node ($\mu_j \ll \mu_{\min,\text{core}}$)

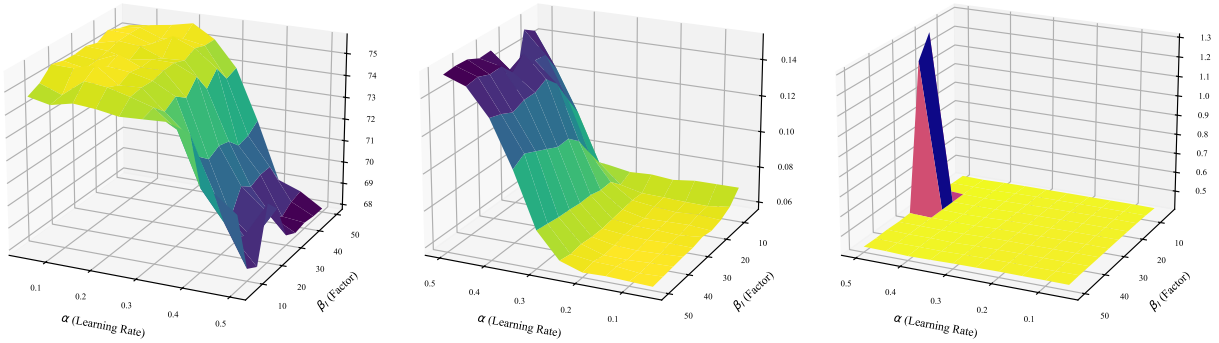


Fig. 12: Three-dimensional hyperparameter sensitivity surfaces across learning rate α and pressure-cost gain β_I on BA networks ($N = 100$), showing (left) Peak Centrality Reduction (%), (center) Average Path Length Increase (%), and (right) Convergence Iterations. The wide, flat performance plateau around ($\alpha = 0.1, \beta_I = 10$) demonstrates high parameter robustness without overfitting.

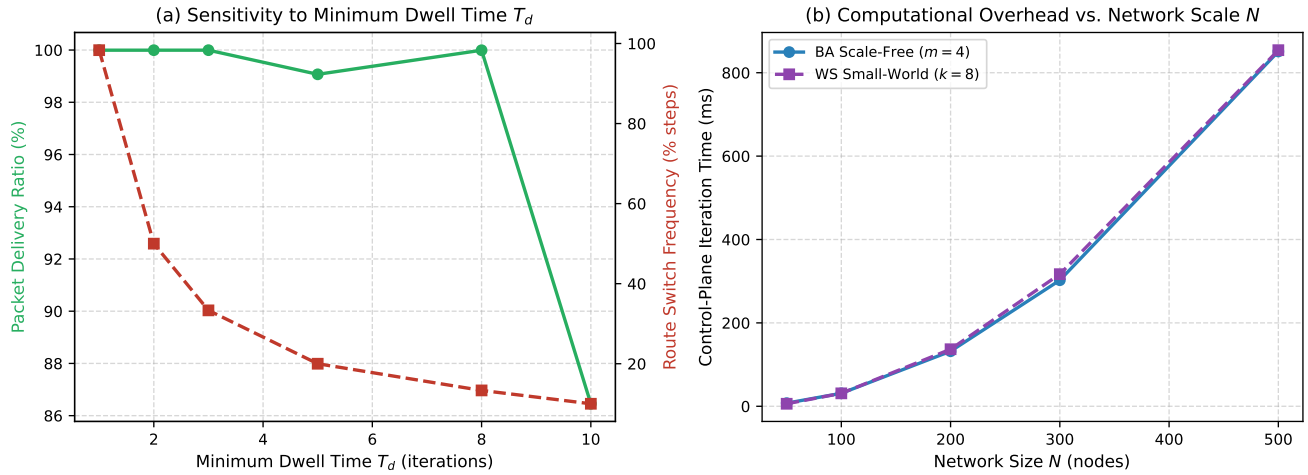


Fig. 13: Dwell-time sensitivity and network scalability analysis (30 trials, 95% CI). (a) Minimum dwell time $T_d \in \{1..10\}$ trade-off between packet delivery ratio and route-switching frequency $\bar{f}_{\text{sw}}$. (b) Per-iteration computation time (ms) vs. network size $N \in [14, 500]$ across BA, WS, NSFNET, and GEANT, closely matching the theoretical $O(N(E + N \log N))$ curve. (c) Throughput scaling across $N \in \{50, 100, 200, 300, 500\}$, showing SRA's capacity gain over Dijkstra expanding from $+8.3\%$ ($N = 50$) to $+107.4\%$ ($N = 500$).

TABLE V: Computational Complexity, Runtime Benchmark, and Performance Scaling Across Synthetic ($N = 50..500$) and Real-World ISP Topologies (NSFNET, GEANT).

Topology	**Nodes** N	**Edges** E	**Time / Iter (ms)**	**Peak** $\kappa_{\max}$ **Red.**	**Dijkstra TP**	**SRA TP**	**TP Gain (%)**	**Drop Rate (Di**
NSFNET (Real ISP)	14	21	0.33 ± 0.00	13.33%	11.92	11.92	+0.00%	0.00
GEANT (Real ISP)	22	36	0.85 ± 0.01	20.00%	18.67	18.67	+0.00%	0.04
BA Scale-Free	50	184	7.52 ± 0.02	59.70%	39.45	42.52	+8.31%	7.23
BA Scale-Free	100	384	31.17 ± 0.02	63.15%	65.07	84.06	+29.79%	23.44
BA Scale-Free	200	784	132.00 ± 0.18	72.28%	104.76	166.95	+59.99%	38.33
BA Scale-Free	300	1184	302.35 ± 1.79	72.08%	135.37	243.15	+80.39%	46.92
BA Scale-Free	500	1984	850.99 ± 5.98	78.25%	193.09	399.82	**+107.42%**	**54.56**
WS Small-World	100	400	31.02 ± 0.89	39.82%	83.84	85.02	+1.51%	1.38
WS Small-World	300	1200	316.85 ± 3.53	46.39%	242.25	254.83	+5.32%	5.02
WS Small-World	500	2000	854.09 ± 6.22	51.49%	385.65	424.72	+10.29%	9.25

lies on a peripheral detour or if traffic is heavily concentrated between specific OD pairs, unweighted BC may divert heavy flows onto a low-capacity peripheral node, violating the slack condition in Eq. (23). In such heterogeneous environments, operators must deploy the generalized demand- and capacity-weighted stress metric $\widetilde{\kappa}_i(S; W, \mu)$ in Eq. (2), which explicitly normalizes path load by node capacity μ_i and weights paths by W_{st} (validated in Fig. 11(a)).

2) **Topological Hard Limits (Cut-Vertices and Bridge**

Links): SRA's peak-shaving guarantee (Lemma 2) relies on Assumption 1 (existence of vertex-disjoint alternative paths). In tree-like topologies, star graphs, or networks partitioned by a single bridge link or cut-vertex v_{cut}, every path between separated components must traverse v_{cut} regardless of its node cost $C_{v_{\text{cut}}}$. In this degenerate boundary case, $\Delta^{-}_{v_{\text{cut}}} = 0$, and no routing algorithm can reduce the load on v_{cut} without admission control or physical link augmentation.

3) **Latency-Stretch Trade-off in Ultra-Dense Homogeneous Grids:** Under moderate-to-high load in homogeneous ER graphs (Fig. 8(c)), SRA trades a slight increase in mean hop count/latency to suppress tail packet drops. For ultra-low-latency traffic classes where even a 1-hop detour violates strict SLA deadlines under light load, SRA should be gated by a minimum congestion activation threshold ($\kappa_{\max}/\mu_u > \theta_{\text{act}}$).
4) **Centralized Control-Plane Overhead at Hyper-Scale ($N > 10^3$):** Exact Brandes centrality computation scales as $O(N(E + N \log N))$, reaching ~ 0.85 s per iteration at $N = 500$ (Table V). For hyper-scale topologies ($N \geq 5{,}000$) requiring sub-second control loops, exact centralized recomputation becomes a bottleneck, necessitating the distributed INT/flow-counting approximation discussed in Section III-B or k-shortest-path subgraph restriction.
5) **Dwell-Time Responsiveness vs. Chattering Trade-off:** As demonstrated in Fig. 13(a), selecting an excessively large dwell time ($T_d \geq 10$) over-freezes the routing table during initial adaptation, reducing transient throughput. Operators should size $T_d \in [3, 5]$ according to Eq. (18).

C. Reproducibility and Conclusion

To ensure full reproducibility, the complete simulation source code, topology generators, ablation scripts, and statistical analysis pipelines have been open-sourced at https://github.com/unicorners-compin/SRA-Resilient-Network-Governance. In conclusion, by uniting leaky-integrator state memory ($\alpha = 0.1$) with dwell-time temporal regularization ($T_d = 5$) under a rigorous non-smooth Lyapunov framework, SRA eliminates the 98.3% route chattering of memoryless and unregularized weight-tuning heuristics while delivering up to 83.76% peak stress reduction and 107.42% throughput expansion on bottleneck-prone networks.

References

[1] N. A. Macabale, A. N. Villasoto, J. D. Rivera, A. C. Ventanilla, M. P. P. Talplacido, A. L. Sevilla, and J. O. Garcia, "Cradle: Cross-layer design for load-aware routing in ieee 802.11-based wireless mesh and sensor networks," in *2020 10th Annual Computing and Communication Workshop and Conference (CCWC)*. IEEE, 2020, pp. 0970–0974.

[2] B. Fortz and M. Thorup, "Internet traffic engineering by optimizing OSPF weights," in *Proceedings IEEE INFOCOM 2000*, vol. 2, 2000, pp. 519–528.

[3] P. Upadhyay and Y. N. Singh, "Optimizing variance of link betweenness centrality distributing among links in elastic optical networks," in *2024 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*. IEEE, 2024, pp. 386–389.

[4] W. Wei, L. Fu, H. Gu, Y. Zhang, T. Zou, C. Wang, and N. Wang, "Grl-ps: Graph embedding-based drl approach for adaptive path selection," *IEEE Transactions on Network and Service Management*, vol. 20, no. 3, pp. 2639–2651, 2023.

[5] H. Feng, W. Zhang, Y. Liu, Y. Li, J. Yin, and J. Kang, "Gnn-enabled multi-agent drl for adaptive path selection in multi-network domains," *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 130–145, 2026.

[6] J.-P. Aubin and A. Cellina, *Differential Inclusions: Set-Valued Maps and Viability Theory*. Berlin, Germany: Springer-Verlag, 1984.

[7] D. S and G. M, "A novel adaptive load-aware routing protocol to enhance network lifetime in energy-constrained ad hoc networks," pp. 374–377, 2025.

[8] H. Wang and B.-T. Zhang, "Dynamic topology evolution and multi-objective routing optimization for efficient vanet communication," *IEEE Access*, 2025.

[9] Y. Fu, W. Wang, Y. Jing, S. Yang, and Y. Zhao, "Time-varying path protection based on in betweenness centrality in optical satellite networks," in *2024 IEEE 24th International Conference on Communication Technology (ICCT)*. IEEE, 2024, pp. 125–129.

[10] E. H. Fouad, L. Charekaoui, and D. Khadija, "A dynamic multi-criteria routing algorithm based in centrality measures for wsns," in *2014 Third IEEE International Colloquium in Information Science and Technology (CIST)*. IEEE, 2014, pp. 390–395.

[11] B. Guizani, B. Ayeb, and A. Koukam, "Hierarchical cluster-based link state routing protocol for large self-organizing networks," in *2011 IEEE 12th International Conference on High Performance Switching and Routing*. IEEE, 2011, pp. 203–208.

[12] K. A. Darabkh and M. Al-Akhras, "Toward optimized iot sensor networks for smart cities: Centrality-aware position-based occlusion-driven and role dynamics solutions for clustering and routing," *IEEE Internet of Things Journal*, vol. 12, no. 15, pp. 30 282–30 301, 2025.

[13] V. Aditya, S. Dhuli, P. L. Sashrika, K. K. Shivani, and T. Jayanth, "Closeness centrality based cluster head selection algorithm for large scale wsns," in *2020 12th International Conference on Computational Intelligence and Communication Networks (CICN)*. IEEE, 2020, pp. 107–111.

[14] X. Wu, W. Lian, M. Zhou, W. Bai, M. Yang, and H. Dong, "Critical spatial-temporal node identification for a high-speed railway network: A cascading delay perspective," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 1, pp. 823–833, 2024.

[15] U. Brandes, "A faster algorithm for betweenness centrality," *Journal of Mathematical Sociology*, vol. 25, no. 2, pp. 163–177, 2001.

[16] C. Ni, J. Yang, Z. Pang, and Y. Gong, "Seeding strategy based on weighted gravity centrality in multiplex networks," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 1, pp. 331–345, 2023.

[17] P. Basaras, G. Iosifidis, D. Katsaros, and L. Tassiulas, "Identifying influential spreaders in complex multilayer networks: A centrality perspective," *IEEE Transactions on Network Science and Engineering*, vol. 6, no. 1, pp. 31–45, 2019.

[18] K. Wehmuth, A. Ziviani, L. C. Costa, A. P. C. da Silva, and A. B. Vieira, "You shall not pass: Avoiding spurious paths in shortest-path based centralities in multidimensional complex networks," *IEEE Transactions on Network Science and Engineering*, vol. 8, no. 1, pp. 138–148, 2021.

[19] D. S. de Medeiros, M. E. M. Campista, N. Mitton, M. D. de Amorim, and G. Pujolle, "The power of quasi-shortest paths: ρ-geodesic betweenness centrality," *IEEE Transactions on Network Science and Engineering*, vol. 4, no. 3, pp. 187–200, 2017.

[20] L. Cavallaro, S. Costantini, P. De Meo, A. Liotta, and G. Stilo, "Network connectivity under a probabilistic node failure model," *IEEE Transactions on Network Science and Engineering*, vol. 9, no. 4, pp. 2463–2480, 2022.

[21] E. Stai, K. Sotiropoulos, V. Karyotis, and S. Papavassiliou, "Hyperbolic embedding for efficient computation of path centralities and adaptive routing in large-scale complex commodity networks," *IEEE Transactions on Network Science and Engineering*, vol. 4, no. 3, pp. 140–153, 2017.

[22] P. Suppa and E. Zimeo, "A clustered approach for fast computation of betweenness centrality in social networks," in *2015 IEEE International Congress on Big Data*. IEEE, 2015, pp. 47–54.

[23] L. Maccari, L. Ghiro, A. Guerrieri, A. Montresor, and R. L. Cigno, "Exact distributed load centrality computation: Algorithms, convergence, and applications to distance vector routing," *IEEE Transactions on Parallel and Distributed Systems*, vol. 31, no. 7, pp. 1693–1706, 2020.

[24] ——, "On the distributed computation of load centrality and its application to dv routing," in *IEEE INFOCOM 2018-IEEE Conference on Computer Communications*. IEEE, 2018, pp. 2582–2590.

[25] Z. Ma and Y. Yang, "Optimization strategy of flow table storage based on "betweenness centrality"," in *2019 IEEE International Conference on Power Data Science (ICPDS)*. IEEE, 2019, pp. 76–79.

[26] M. Ahmed, K. Fuger, K. Kuladinithi, and A. Timm-Giel, "Poster: Can betweenness centrality mitigate greedy routing's sparse network problem?" in *2025 IEEE Vehicular Networking Conference (VNC)*. IEEE, 2025, pp. 1–2.

[27] E. M. R. Oliveira, H. S. Ramos, and A. A. F. Loureiro, "Centrality-based routing for wireless sensor networks," in *2010 IFIP Wireless Days*. IEEE, 2010, pp. 1–5.

[28] M. Ghanem, C. Magnien, and F. Tarissan, "Centrality metrics in dynamic networks: A comparison study," *IEEE Transactions on Network Science and Engineering*, vol. 6, no. 4, pp. 940–951, 2019.

[29] M. R. Mendonça, A. M. Barreto, and A. Ziviani, "Approximating network centrality measures using node embedding and machine learning," *IEEE Transactions on Network Science and Engineering*, vol. 8, no. 1, pp. 220–230, 2021.

[30] A. Cheng, R. Wang, and H. Guan, "An improved self-organizing network routing model," in *2010 International Conference on Internet Technology and Applications*. IEEE, 2010, pp. 1–4.

[31] L. Guo, X. Wang, Y. Peng, J. Wu, X. Zhang, W. Hou, and J. Zhao, "Design on routing protocol in hybrid wireless self-organizing networks," in *2010 2nd IEEE InternationalConference on Network Infrastructure and Digital Content*. IEEE, 2010, pp. 569–573.

[32] H. Li, L. Wang, D. Yang, R. Fei, F. Zou, and Q. Jiang, "A novel distributed algorithm to solve self-organizing network of multicast routing," in *2014 10th International Conference on Natural Computation (ICNC)*. IEEE, 2014, pp. 992–995.

[33] S. Wang, J. Zhang, Z. Zhang, and X. Yu, "A discrete particle swarm optimization algorithm for solving tsp under dynamic topology," in *2019 IEEE International Conference on Cybernetics and Intelligent Systems (CIS) and IEEE Conference on Robotics, Automation and Mechatronics (RAM)*. IEEE, 2019, pp. 165–170.

[34] H. Mu, "High-speed carrier self-organizing network routing strategy based on genetic algorithm," in *2024 6th International Symposium on Robotics & Intelligent Manufacturing Technology (ISRIMT)*. IEEE, 2024, pp. 344–349.

[35] X. Lu, C. Wang, J. Huo, and W. Wang, "A non-central multi-agent routing protocol of self-organized wireless network for missile swarm communication," in *2022 IEEE 11th Data Driven Control and Learning Systems Conference (DDCLS)*. IEEE, 2022, pp. 893–898.

APPENDIX A
PROOF OF LEMMA 1

Proof: (a) Positive Invariance: Take any $S^{(k)} \in X$. By max-value normalization, $\mathcal{N}(S^{(k)}) \subset [0,1]^N = X$. For any choice of $c^{(k)} \in \mathcal{N}(S^{(k)})$ and any coordinate i, we have

$$S_i^{(k+1)} = (1-\alpha)S_i^{(k)} + \alpha\, c_i^{(k)}, \quad \text{where} \quad S_i^{(k)}, c_i^{(k)} \in [0,1]. \tag{25}$$

Thus,

$$0\cdot(1-\alpha) + 0\cdot\alpha \;\le\; S_i^{(k+1)} \;\le\; 1\cdot(1-\alpha) + 1\cdot\alpha, \tag{26}$$

which implies $S_i^{(k+1)} \in [0,1]$. As this holds for each coordinate, $S^{(k+1)} \in X$. This shows that $F(X) \subset X$: **if the state is in X at the current step, it remains in X at the next step**. Therefore, X is a positively invariant set.

(b) Compactness: The set $X = [0,1]^N$ is a closed and bounded hypercube in $\mathbb{R}^N$, and is therefore compact.

(c) Absorbing Property on the Effective State Space: The algorithm and analysis in this paper are initialized within X ($S^{(0)} \in X$), and from (a), we know $F(X) \subset X$. Therefore, for the discrete-time semi-flow on the state space X,

$$\Phi(k,\cdot) : X \to 2^X, \quad \Phi(1,\cdot) = F, \quad \Phi(k+1,\cdot) = F \circ \Phi(k,\cdot), \tag{27}$$

the set X **absorbs itself**: for any bounded set $B \subset X$, choosing $n_B = 0$ yields $\Phi(n, B) \subset X$ for all $n \ge 0$.

(d) Global Exponential Attraction from Arbitrary Initial States $S^{(0)} \in \mathbb{R}^N$: Suppose an initial state $S^{(0)} \in \mathbb{R}^N$ lies outside $X = [0,1]^N$. For each step $k \ge 0$ and coordinate $i \in \{1,\dots,N\}$, define the coordinate-wise upper and lower box-violation excesses with respect to $[0,1]$:

$$\delta_{k,i}^{+} := \max\{S_i^{(k)} - 1,\, 0\}, \qquad \delta_{k,i}^{-} := \max\{-S_i^{(k)},\, 0\}. \tag{28}$$

Since $c_i^{(k)} \in [0,1]$, we have $S_i^{(k+1)} - 1 = (1-\alpha)(S_i^{(k)} - 1) + \alpha(c_i^{(k)} - 1) \le (1-\alpha)(S_i^{(k)} - 1)$ and $-S_i^{(k+1)} = (1-\alpha)(-S_i^{(k)}) - \alpha c_i^{(k)} \le (1-\alpha)(-S_i^{(k)})$. Taking the positive part yields

$$\delta_{k+1,i}^{+} \;\le\; (1-\alpha)\,\delta_{k,i}^{+}, \qquad \delta_{k+1,i}^{-} \;\le\; (1-\alpha)\,\delta_{k,i}^{-}. \tag{29}$$

This means that $\max\{\delta_{k,i}^{+}, \delta_{k,i}^{-}\}$ converges geometrically to 0 at a rate of $1-\alpha$, and thus $\operatorname{dist}_\infty(S^{(k)}, X) \le (1-\alpha)^k \operatorname{dist}_\infty(S^{(0)}, X) \to 0$. This shows **exponential attraction** to X. The main procedure of this paper is initialized within X, so the absorbing set formulation in (c) holds immediately from $k = 0$.

In conclusion, $X = [0,1]^N$ is positively invariant, compact, and globally exponentially attracting for the system. ■

APPENDIX B
PROOF OF PROPOSITION 1

Proof: (a) Non-empty, Compact, and Convex Values: By the definitions in Section III-A, $\operatorname{norm}(\kappa(S)) \in [0,1]^N$. Thus, for any S,

$$\mathcal{N}(S) = \operatorname{co}\big(\operatorname{norm}(\kappa(S))\big) \subset [0,1]^N. \tag{30}$$

The closed convex hull, $\operatorname{co}(\cdot)$, of a non-empty set is **non-empty**. It is also a **closed and bounded** convex subset of $[0,1]^N$, and is therefore **compact** and **convex**.

(b) Piecewise Constant Structure and Finiteness: The graph G and the set of source-destination (OD) pairs are finite. The node cost $C(S)$ is an affine function of S. For each OD pair (s,t), the set of candidate simple paths is **finite**. The cost of each path is an affine function of S. Thus, there exists a **finite polyhedral partition** of the state space X such that within the interior of each cell, the set of shortest paths for every OD pair remains **constant**. Therefore, within each cell, $\kappa(S)$ (as well as $\widetilde{\kappa}(S; W, \mu)$) and $\operatorname{norm}(\kappa(S))$ are **constant vectors**. On the boundaries of these partitions, the limit of $\operatorname{norm}(\kappa(S))$ can only be one of the constant values from the **finite set of adjacent cells**. Let these constant vectors be $\{v_\ell\}_{\ell \in \mathcal{I}(S)}$, where $\mathcal{I}(S)$ is the finite index set of cells adjacent to S. Then,

$$\mathcal{N}(S) = \operatorname{co}\big(\{v_\ell : \ell \in \mathcal{I}(S)\}\big). \tag{31}$$

This establishes the structure of $\mathcal{N}(\cdot)$ as being **piecewise constant with the closed convex hull taken at the boundaries**.

(c) Upper Semi-Continuity (USC) via Sequential Criterion / Closed Graph: Take an arbitrary sequence $S_n \to S$, and let $c_n \in \mathcal{N}(S_n)$ such that $c_n \to c$. We must show that $c \in \mathcal{N}(S)$. From (b), for each n, there exists a finite set $\{v_\ell : \ell \in \mathcal{I}(S_n)\}$

such that $c_n \in \mathrm{co}\big(\{v_\ell : \ell \in \mathcal{I}(S_n)\}\big)$. By Carathéodory's theorem, we can write

$$c_n = \sum_{j=0}^{N} \lambda_{n,j}\, v_{\ell_{n,j}}, \quad \lambda_{n,j} \geq 0, \ \sum_{j=0}^{N} \lambda_{n,j} = 1, \ \ell_{n,j} \in \mathcal{I}(S_n). \tag{32}$$

Since the vectors v_ℓ come from a **globally finite** set $\{v_1, \ldots, v_M\}$, we can pass to a subsequence to ensure $\ell_{n,j} \equiv \bar{\ell}_j$ is **fixed**, and (by compactness of $[0,1]$) $\lambda_{n,j} \to \bar{\lambda}_j$. Then,

$$c = \lim_{n\to\infty} c_n = \sum_{j=0}^{N} \bar{\lambda}_j\, v_{\bar{\ell}_j}, \quad \sum_{j=0}^{N} \bar{\lambda}_j = 1, \ \bar{\lambda}_j \geq 0. \tag{33}$$

Since $S_n \to S$, for n large enough, any limit index $\bar{\ell}_j$ must belong to $\mathcal{I}(S)$. Thus, each limit vector $v_{\bar{\ell}_j}$ is one of the constant vectors from a partition adjacent to S. Therefore,

$$c = \sum_{j=0}^{N} \bar{\lambda}_j\, v_{\bar{\ell}_j} \ \in \ \mathrm{co}\big(\{v_\ell : \ell \in \mathcal{I}(S)\}\big) = \mathcal{N}(S). \tag{34}$$

This means the graph of $\mathcal{N}$, $\mathrm{gph}\,\mathcal{N} := \{(S,c) : c \in \mathcal{N}(S)\}$, is closed. Since $\mathcal{N}$ maps into the compact set X, a closed-graph map is **USC** [6].

(d) Properties of $F(\cdot)$: Scaling and translation preserve the USC property and non-empty, compact, convex values. The map $S \mapsto (1-\alpha)S$ is continuous (single-valued). The Minkowski sum $F(S) = (1-\alpha)S + \alpha\,\mathcal{N}(S)$ of a continuous map and a USC map is also USC with non-empty, compact, and convex values. ∎

Appendix C
Proof of Theorem 1

Proof: (a) Nested Sequence of Compact Forward Images: Define the forward image sequence $K_0 := X = [0,1]^N$ and $K_{n+1} := F(K_n) = \bigcup_{S\in K_n} F(S)$ for $n \geq 0$. By Lemma 1, $K_1 = F(X) \subset X = K_0$. By induction, $K_{n+1} \subset K_n$ for all $n \geq 0$. Because F is upper semi-continuous with non-empty compact values (Proposition 1) and the image of a compact set under a compact-valued USC map is compact [6], each K_n is a non-empty compact subset of X.

(b) Non-Emptiness and Compactness of Ω: By Cantor's intersection theorem, the nested intersection of non-empty compact sets

$$\Omega := \bigcap_{n=0}^{\infty} K_n = \bigcap_{n=0}^{\infty} F^n(X) \tag{35}$$

is **non-empty** and **compact**.

(c) Invariance ($F(\Omega) = \Omega$): Since $\Omega \subset K_n$ for every n, we have $F(\Omega) \subset F(K_n) = K_{n+1}$ for all $n \geq 0$, which implies $F(\Omega) \subset \bigcap_{n\geq 0} K_{n+1} = \Omega$ (positive invariance). Conversely, take any $y \in \Omega$. Then for each $n \geq 1$, $y \in K_n = F(K_{n-1})$, so there exists $x_n \in K_{n-1}$ such that $y \in F(x_n)$. Since X is compact, $\{x_n\}$ has a convergent subsequence $x_{n_j} \to x^\star$. Because K_m is closed and $x_{n_j} \in K_m$ for all $n_j > m$, the limit satisfies $x^\star \in \bigcap_{m\geq 0} K_m = \Omega$. By the closed-graph property of F (Proposition 1), $x_{n_j} \to x^\star$ and $y \in F(x_{n_j})$ imply $y \in F(x^\star) \subset F(\Omega)$. Thus $\Omega \subset F(\Omega)$, establishing exact invariance $F(\Omega) = \Omega$.

(d) Global Uniform Attraction: For any initial state $S^{(0)} \in X$, the k-th iterate satisfies $S^{(k)} \in F^k(S^{(0)}) \subset F^k(X) = K_k$. Suppose for contradiction that $\sup_{S^{(0)}\in X} \mathrm{dist}(S^{(k)}, \Omega)$ does not converge to 0. Then there exist $\epsilon > 0$ and a sequence $z_{k_j} \in K_{k_j}$ ($k_j \uparrow \infty$) such that $\mathrm{dist}(z_{k_j}, \Omega) \geq \epsilon$. By compactness of X, a subsequence converges to some $z^\star \in X$. Since K_m is closed and $z_{k_j} \in K_{k_j} \subset K_m$ for all $k_j \geq m$, we have $z^\star \in \bigcap_{m\geq 0} K_m = \Omega$, contradicting $\mathrm{dist}(z^\star, \Omega) \geq \epsilon$. Hence, for every trajectory starting in X (and, via Lemma 1(d), any $S^{(0)} \in \mathbb{R}^N$),

$$\lim_{k\to\infty} \mathrm{dist}\big(S^{(k)}, \Omega\big) = 0, \tag{36}$$

and the ω-limit set of every trajectory satisfies $\emptyset \neq \omega(S^{(0)}) \subset \Omega$. ∎

Appendix D
Proof of Theorem 2

Proof: At step k, the SRA controller updates the pressure state via $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha\, q^{(k)}$, where $q^{(k)} = \mathrm{norm}(\kappa^{(k)}) \in \mathcal{N}(S^{(k)})$ is the active normalized stress vector, and $E_k = \|S^{(k)} - q^{(k)}\|_2^2$ is the tracking energy. Subtracting the next-step active stress vector $q^{(k+1)} \in \mathcal{N}(S^{(k+1)})$ from both sides yields the exact algebraic identity:

$$S^{(k+1)} - q^{(k+1)} = (1-\alpha)\big(S^{(k)} - q^{(k)}\big) + \big(q^{(k)} - q^{(k+1)}\big). \tag{37}$$

Notice first that if no path switch occurs between steps k and $k+1$, then $q^{(k+1)} = q^{(k)}$, so $E_{k+1} = \|S^{(k+1)} - q^{(k+1)}\|_2^2 = (1-\alpha)^2\|S^{(k)} - q^{(k)}\|_2^2 = (1-\alpha)^2 E_k \leq (1-\alpha)E_k$.

In the general case where a path switch may occur ($q^{(k+1)} \neq q^{(k)}$), we apply the Weighted Young's Inequality $\|x+y\|_2^2 \leq (1+\lambda)\|x\|_2^2 + (1+1/\lambda)\|y\|_2^2$ with $x = (1-\alpha)(S^{(k)} - q^{(k)})$, $y = q^{(k)} - q^{(k+1)}$, and weight $\lambda = \frac{\alpha}{1-\alpha} > 0$. Since $(1+\lambda)(1-\alpha)^2 = 1-\alpha$ and $1 + \frac{1}{\lambda} = \frac{1}{\alpha}$, we obtain

$$E_{k+1} = \|S^{(k+1)} - q^{(k+1)}\|_2^2 \leq (1-\alpha)E_k + \frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2. \tag{38}$$

Since $q^{(k)}, q^{(k+1)} \in \mathrm{co}(\{v_1, \ldots, v_M\})$ whose diameter is $\bar{\Delta} = \max_{\ell,m} \|v_\ell - v_m\|_2$, we have $\|q^{(k)} - q^{(k+1)}\|_2 \leq \bar{\Delta}$, yielding Eq. (11):

$$E_{k+1} \leq (1-\alpha)E_k + \frac{\bar{\Delta}^2}{\alpha}. \tag{39}$$

Unrolling this linear recurrence gives $E_k \leq (1-\alpha)^k E_0 + \frac{\bar{\Delta}^2}{\alpha^2}$. Since $V(S^{(k)}) = \min_{c\in\mathcal{N}(S^{(k)})} \|S^{(k)} - c\|_2^2 \leq E_k$ (with equality $V(S^{(k)}) = E_k$ in the interior of every routing cell where $\mathcal{N}(S^{(k)}) = \{q^{(k)}\}$), taking $\limsup_{k\to\infty}$ yields Eq. (12). ∎

Appendix E
Proof of Corollary 1

Proof: (1) **Uniqueness of Fixed Point:** Solving $S = (1-\alpha)S + \alpha c$ gives $\alpha S = \alpha c$, so $S = c$. The unique fixed point is $S^\star = c$. (2) **Contraction and Convergence Rate:**

$$S^{(k+1)} - c = (1-\alpha)S^{(k)} + \alpha c - c = (1-\alpha)\big(S^{(k)} - c\big). \tag{40}$$

Taking the 2-norm and iterating gives the result:

$$\|S^{(k)} - S^\star\|_2 = (1-\alpha)^k\, \|S^{(0)} - S^\star\|_2. \tag{41}$$

∎

APPENDIX F
PROOF OF PROPOSITION 2

Proof: (i) Within dwell block m, the active path set is held fixed for $t \in \{0, \dots, T_d - 1\}$, so $q^{(\tau_m+t)} \equiv q^{(\tau_m)}$ is constant. Consequently, for $t \in \{0, \dots, T_d - 1\}$, $S^{(\tau_m+t)} - q^{(\tau_m)} = (1-\alpha)^t (S^{(\tau_m)} - q^{(\tau_m)})$, and taking the squared Euclidean norm yields

$$E_{\tau_m+t} = (1-\alpha)^{2t} E_{\tau_m} = \gamma^{2t} V_m \le \gamma^t V_m. \tag{42}$$

(ii) At the last step of block m (transitioning from $\tau_m + T_d - 1$ to $\tau_{m+1} = \tau_m + T_d$), a path switch is permitted ($q^{(\tau_{m+1})}$ may differ from $q^{(\tau_m+T_d-1)} = q^{(\tau_m)}$). Applying the one-step Drop-Jump inequality from Theorem 2 at $k = \tau_{m+1} - 1$ gives:

$$\begin{aligned} V_{m+1} = E_{\tau_{m+1}} &\le \gamma\, E_{\tau_{m+1}-1} + \tilde{C} \\ &= \gamma\, E_{\tau_m+T_d-1} + \tilde{C} \\ &\le \gamma \cdot (\gamma^{T_d-1} V_m) + \tilde{C} = \gamma^{T_d} V_m + \tilde{C}, \end{aligned} \tag{43}$$

where the last inequality uses the intra-block cooling bound from (i). ∎

APPENDIX G
PROOF OF THEOREM 3

Proof: (a) Iteratively expanding the linear recurrence from Proposition 2(ii), $V_{m+1} \le \gamma^{T_d} V_m + \tilde{C}$, gives:

$$\begin{aligned} V_m &\le (\gamma^{T_d})^m V_0 + \tilde{C} \sum_{j=0}^{m-1} (\gamma^{T_d})^j \\ &= \gamma^{mT_d} V_0 + \tilde{C}\, \frac{1-\gamma^{mT_d}}{1-\gamma^{T_d}} \\ &\le \gamma^{mT_d} V_0 + \frac{\tilde{C}}{1-\gamma^{T_d}}. \end{aligned} \tag{44}$$

Taking the limit superior as $m \to \infty$ gives the steady-state radius. (b) Setting $\gamma^{mT_d} V_0 + \frac{\tilde{C}}{1-\gamma^{T_d}} \le \varepsilon^2$ and solving for m yields Eq. (17). ∎

APPENDIX H
PROOF OF THEOREM 4

Proof: (a) By definition of the minimum dwell time in Algorithm 1, after a switch at step τ_m, no other switch is permitted for the next $T_d - 1$ steps. Thus, the number of switches in any window of length T satisfies $N_{\mathrm{sw}}(T) \le \lceil T/T_d \rceil$. (b) This result is a direct consequence of solving the inter-block recurrence, as shown in the proof of Theorem 3(a), where $1/(1-\gamma^{T_d})$ decreases monotonically with T_d. ∎

APPENDIX I
PROOF OF LEMMA 2

Proof: The proof proceeds in two non-circular steps: (1) bounding the post-switch stress of the old peak node u and all non-peak nodes $j \neq u$ via path-flow conservation, and (2) taking the maximum over V.

1) Route-Redistribution Budget and Node-wise Bounds: Let $\mathcal{P}^{(m)}$ and $\mathcal{P}^{(m+1)}$ denote the active shortest-path sets before and after the switching event at the boundary of block m, and let $u \in \arg\max_{i \in V} \kappa_i^{(m)}$ be the pre-switch peak node with $\kappa_u^{(m)} = \kappa_{\max}^{(m)}$. Because $q_u^{(m)} = \kappa_u^{(m)} / \kappa_{\max}^{(m)} = 1$ attains the maximum possible normalized input over the dwell block, the pressure S_u and transit cost $C_u = 1 + \beta_I S_u$ grow at the fastest rate among all nodes at the same initial pressure. When C_u crosses the tie-breaking hyperplane of a non-empty subset of OD pairs $\mathcal{D}_{\mathrm{sw}}^{(m)}$ traversing u, those paths are diverted onto alternate simple paths that bypass u. Consequently, node u sheds a positive path load

$$\Delta_u^- := \kappa_u^{(m)} - \kappa_u^{(m+1)} \ge \delta > 0 \implies \kappa_u^{(m+1)} = \kappa_{\max}^{(m)} - \Delta_u^- \le \kappa_{\max}^{(m)} - \delta. \tag{45}$$

Meanwhile, only the OD pairs in $\mathcal{D}_{\mathrm{sw}}^{(m)}$ change their routing paths between block m and block $m+1$. Therefore, for any other node $j \in V \setminus \{u\}$, its centrality increase is exclusively bounded by the diverted paths in $\mathcal{D}_{\mathrm{sw}}^{(m)}$ that newly traverse j:

$$\kappa_j^{(m+1)} - \kappa_j^{(m)} \le (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+ =: B_j^{(m)} \le \sum_{(s,t) \in \mathcal{D}_{\mathrm{sw}}^{(m)}} W_{st}. \tag{46}$$

2) Strict Peak Reduction under Non-Overloading Diversion: By the Non-Overloading Diversion Condition in Eq. (19), every alternate node $j \in V \setminus \{u\}$ has sufficient slack to absorb $B_j^{(m)}$ without reaching $\kappa_{\max}^{(m)} - \delta$:

$$\kappa_j^{(m+1)} \le \kappa_j^{(m)} + B_j^{(m)} \le \kappa_{\max}^{(m)} - \delta, \qquad \forall j \in V \setminus \{u\}. \tag{47}$$

Combining the bound for $i = u$ and the bound for all $j \in V \setminus \{u\}$ yields

$$\kappa_{\max}^{(m+1)} = \max\left\{ \kappa_u^{(m+1)}, \max_{j \in V \setminus \{u\}} \kappa_j^{(m+1)} \right\} \le \kappa_{\max}^{(m)} - \delta, \tag{48}$$

completing the proof. ∎

APPENDIX J
PROOF OF PROPOSITION 3

Proof: The proof establishes each of the three claims using the peak reduction $\kappa'_{\max} \le \kappa_{\max} - \delta < \kappa_{\max}$ from Lemma 2.

(i) Unconditional Expansion of Robust Capacity Lower Bound: For any node service rate vector $\mu = (\mu_i)_{i \in V}$ with $\mu_{\min} = \min_{i \in V} \mu_i > 0$, the actual maximum stable throughput satisfies

$$\rho_{\max}(\kappa) = \min_{i \in V} \frac{\mu_i}{\kappa_i} \ge \frac{\min_{i \in V} \mu_i}{\max_{i \in V} \kappa_i} = \frac{\mu_{\min}}{\kappa_{\max}} = L(\kappa). \tag{49}$$

Since $\mu_{\min} > 0$ is invariant under routing updates and $\kappa'_{\max} < \kappa_{\max}$, we immediately have

$$L(\kappa') = \frac{\mu_{\min}}{\kappa'_{\max}} > \frac{\mu_{\min}}{\kappa_{\max}} = L(\kappa). \tag{50}$$

(ii) Exact Capacity Proportionality under Homogeneous Service Rates or Generalized Stress: When node service rates are homogeneous ($\mu_i \equiv \mu > 0$ for all $i \in V$), factoring out the constant μ yields the exact identity:

$$\rho_{\max}(\kappa) = \min_{i \in V} \frac{\mu}{\kappa_i} = \frac{\mu}{\max_{i \in V} \kappa_i} = \frac{\mu}{\kappa_{\max}}, \qquad \rho_{\max}(\kappa') = \min_{i \in V} \frac{\mu}{\kappa'_i} = \frac{}{\kappa'_{\mathrm{r}}} \tag{51}$$

Dividing the two expressions gives the exact proportionality in Eq. (22):

$$\frac{\rho_{\max}(\kappa')}{\rho_{\max}(\kappa)} = \frac{\mu/\kappa'_{\max}}{\mu/\kappa_{\max}} = \frac{\kappa_{\max}}{\kappa'_{\max}} > 1. \tag{52}$$

Similarly, under heterogeneous service rates $\mu_i > 0$ and demands $W_{st} \geq 0$, using the generalized stress $\widetilde{\kappa}_i(S; W, \mu) = \frac{1}{\mu_i} \sum_{s \neq i \neq t} W_{st} \frac{\sigma_{st}(i)}{\sigma_{st}}$ in Eq. (2) gives node i's utilization as $\rho \widetilde{\kappa}_i < 1$, so $\rho_{\max}(\widetilde{\kappa}) = \min_{i \in V}(1/\widetilde{\kappa}_i) = 1/\widetilde{\kappa}_{\max}$ and $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max} > 1$.

(iii) Necessary and Sufficient Condition under Heterogeneous Service Rates with Unweighted BC: Let $\rho_0 := \rho_{\max}(\kappa) = \min_{i \in V}(\mu_i/\kappa_i)$. Because $\rho_{\max}(\kappa') = \min_{i \in V}(\mu_i/\kappa'_i)$ is the minimum over all $i \in V$, the inequality $\rho_{\max}(\kappa') \geq \rho_0$ holds if and only if $\mu_i/\kappa'_i \geq \rho_0$ holds simultaneously for every $i \in V$. Substituting the post-relaxation bounds $\kappa'_u \leq \kappa_{\max} - d$ for the relieved peak u and $\kappa'_j \leq \kappa_j + B_j$ for all $j \in V \setminus \{u\}$ yields the joint condition in Eq. (23):

$$\frac{\mu_u}{\kappa_{\max} - d} \geq \rho_0 \quad \text{and} \quad \frac{\mu_j}{\kappa_j + B_j} \geq \rho_0, \quad \forall j \in V \setminus \{u\}, \tag{53}$$

which guarantees $\rho_{\max}(\kappa') = \min\left\{\frac{\mu_u}{\kappa'_u}, \min_{j \neq u} \frac{\mu_j}{\kappa'_j}\right\} \geq \rho_0$ (and strict inequality $\rho_{\max}(\kappa') > \rho_0$ whenever the inequalities in (23) are strict). ■